

A Universal Field-Size Cap and Prize-Facing Completion Program for Mutual Correlated Agreement on Smooth Reed–Solomon Domains

Version 10: quotient, extension-pole, and Hankel rank-drop certificates

Przemek Chojceki

ulam.ai

July 1, 2026

v10 integrated milestone upgrade

Abstract

We prove a field-size-universal upper bound on the proximity parameter achievable in the grand mutual-correlated-agreement (MCA) challenge of Arnon–Boneh–Fenzi [ABF26]. Let $\mathbb{F}$ be any finite field with $|\mathbb{F}| < 2^{256}$, let D be a smooth multiplicative evaluation domain of order n contained in a subfield $B \subseteq \mathbb{F}$ (with $B = \mathbb{F}$ allowed), and let $C = \text{RS}[\mathbb{F}, D, k]$ have $k = \rho n \leq 2^{40}$ and $\rho \in \{1/2, 1/4, 1/8, 1/16\}$. If $2^{10} \mid n$ for $\rho \in \{1/2, 1/4, 1/8\}$, and $2^{11} \mid n$ for $\rho = 1/16$, then

$$\varepsilon_{\text{mca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{n}{|\mathbb{F}|}\right) \geq 2^{-86} \gg 2^{-128}$$

for every $\delta \in [1 - \rho - 2^{-9}, 1 - \rho)$ at rates $1/2, 1/4, 1/8$, and for every $\delta \in [1 - \rho - 2^{-10}, 1 - \rho)$ at rate $1/16$. The same lower bound holds for $\varepsilon_{\text{ca}}(C, \delta)$ on the integer-admissible subinterval ending at $1 - \rho - 1/n$. If $|\mathbb{F}| \geq 2n$, the displayed lower bound improves to 2^{-42} . Consequently $\delta_C^*(2^{-128}) \leq 1 - \rho - 2^{-9}$ for $\rho \in \{1/2, 1/4, 1/8\}$ and $\delta_C^*(2^{-128}) \leq 1 - \rho - 2^{-10}$ for $\rho = 1/16$, throughout the $|\mathbb{F}| < 2^{256}$ challenge envelope. This is a prize-facing negative result: it does not determine the exact threshold, but it rules out an entire near-capacity band for every challenge-envelope row satisfying the printed divisor hypothesis.

The mechanism composes a pigeonhole lower bound on quotient-locator list fibers—a slack-two and subfield-pigeonhole sharpening of [Cho26a, Main Thm. (d)]—with a self-contained deep-point list-to-correlated-agreement conversion. This conversion is the same shallow consequence previously imported from Crites–Stewart [CS25, Thm. 2], but it is proved here directly with the integer-radius condition. The composition itself was first carried out, conditionally and with weaker constants, in the universal-cap theorem of [Cho26b] (error 2^{-42} at gap 2^{-11} , assuming $|\mathbb{F}| \geq 2n$); the present note sharpens the gap, removes the field-size condition, and extends the cap to extension fields. No value-set or equidistribution input is used, so the per-fiber collision problem isolated in [Cho26b] is routed around rather than solved. Version 10 adds a quantitative inverse form of the deep-point conversion, heaviest quotient-remainder prefix fibers, an extension-pole conversion producing genuinely extension-valued witness lines, an exact divisor-block coefficient formula for the support-union ledger, a gcd/lcm quotient-image ledger that coalesces duplicate finite, projective, and curve parameters inside declared quotient branches, and a canonical maximal-minor gcd ledger for regular overdetermined Hankel buckets. We instantiate the bound at the

KoalaBear-sextic parameters of [ABF26, §6.3], including the interleaved rows of their Tables 2–3, give an independent slacked variant through [BCHKS25, Thm. 1.9], and prove a two-radius refinement of the subfield-confinement theorem of [Cho26b], showing that over extension fields every explicit base-rational locator line is inert while the extension-pole floor gives a concrete source of genuinely $\mathbb{F}$ -valued certifying lines whenever a large list is present.

1 Introduction

Proximity testing for Reed–Solomon codes underlies the soundness of modern SNARKs, and the sharpest available soundness accounting is phrased through *mutual correlated agreement* (MCA): a random linear combination $f_1 + \gamma f_2$ being δ -close to the code should force f_1, f_2 to be simultaneously explained by codewords on a *common* agreement set, except with small probability $\varepsilon_{\text{mca}}(C, \delta)$ over γ . The survey of Arnon, Boneh and Fenzi [ABF26] crystallizes the state of the art into a *grand MCA challenge*: for $C = \text{RS}[\mathbb{F}, L, k]$ over a smooth domain L , with rate $\rho \in \{1/2, 1/4, 1/8, 1/16\}$, degree $k \leq 2^{40}$, and field size $|\mathbb{F}| < 2^{256}$, determine the largest proximity parameter δ^* at which $\varepsilon_{\text{mca}}(C, \delta) \leq 2^{-128}$ can hold.

Positive results reach the Johnson bound $1 - \sqrt{\rho}$. On the negative side, [Cho26b] showed $\delta_C^*(2^{-128}) < 1 - \rho - 1/64$ for all prime fields of size up to roughly 2^{150} (per-rate reaches between $2^{150.6}$ and $2^{161.8}$), by exhibiting quotient-locator witnesses whose bad-slope sets are value sets of restricted symmetric sums over small subgroups; it further proved a *conditional* universal cap over every field of size at most 2^{256} — error 2^{-42} at gap 2^{-11} , assuming $|\mathbb{F}| \geq 2n$ — by composing quotient-core lists with the same conversion isolated and proved directly here (its universal-cap theorem). Above the value-set reach, what remained open was the *error-one* question, identified there with the *per-fiber collision problem*: controlling, at a *fixed* large prime, the collisions of characteristic-zero locator values under reduction. Extension fields, the setting actually deployed ([ABF26, §6.3] works over a sextic extension of KoalaBear), posed an additional obstruction: by the subfield-confinement theorem of [Cho26b], refined in Section 6, every explicit witness of [Cho26a, Cho26b] is B -rational and contributes only $p^{-5} < 2^{-154}$ over the deployed extension.

This note sharpens the universal cap — gap 2^{-9} (resp. 2^{-10}) in place of 2^{-11} , error 2^{-86} with no condition relating n and $|\mathbb{F}|$, and extension fields handled directly — for every field below 2^{256} at once, by changing what is counted.

Theorem 1.1 (informal; see Theorem 4.1 and Corollary 5.1). *Let $\mathbb{F}$ be any finite field with $|\mathbb{F}| < 2^{256}$, let $D \subseteq \mathbb{F}^\times$ be a smooth multiplicative evaluation domain of order n , and let $C = \text{RS}[\mathbb{F}, D, k]$ have $k = \rho n \leq 2^{40}$ with $\rho \in \{1/2, 1/4, 1/8, 1/16\}$. Assume $2^{10} \mid n$ for $\rho \in \{1/2, 1/4, 1/8\}$, and $2^{11} \mid n$ for $\rho = 1/16$. Then*

$$\varepsilon_{\text{mca}}(C, \delta) > 2^{-86} \gg 2^{-128}$$

for every $\delta \in [1 - \rho - 2^{-9}, 1 - \rho]$ at rates $1/2, 1/4, 1/8$, and for every $\delta \in [1 - \rho - 2^{-10}, 1 - \rho]$ at rate $1/16$. The same lower bound holds for $\varepsilon_{\text{ca}}(C, \delta)$ on the integer-admissible subinterval ending at $1 - \rho - 1/n$. If $|\mathbb{F}| \geq 2n$, the lower bound is $\geq 2^{-42}$. Hence $\delta_C^(2^{-128}) \leq 1 - \rho - 2^{-9}$ for $\rho \in \{1/2, 1/4, 1/8\}$ and $\delta_C^*(2^{-128}) \leq 1 - \rho - 2^{-10}$ for $\rho = 1/16$.*

The proof composes two ingredients. The first (Lemma 3.1) is a pigeonhole lower bound on list sizes at a single received word: the quotient locators of [Cho26a] attach to each ℓ -subset A of the order- N quotient coset $Q = D^a$ a codeword c_A at distance $\leq 1 - \rho - t/N$ from the word $u_{z_A} = x^{k+ta} + z_A x^{k+(t-1)a}$, where $z_A = -e_1(A)$; pigeonholing the $\binom{N}{\ell}$ subsets over their $\leq |B|$ slope values produces one word u_z carrying a list of size $\binom{N}{\ell}/|B|$. Two refinements of

[Cho26a, Main Thm. (d)] matter here: we run the construction at *slack two* ($t = 2$), so that the listed codewords have degree $\leq k$ and live in $\text{RS}[\mathbb{F}, D, k + 1]$ without any divisibility demand on $k + 1$ (which is odd in all challenge instantiations, while a is a 2-power); and we pigeonhole over the *subfield* B containing D rather than over $\mathbb{F}$, which is what makes the extension-field case go through. The second ingredient is the direct deep-point conversion of Theorem 2.6: in the integer-error range $f \leq n - k - 1$, if $\varepsilon_{\text{ca}}(C, \delta)$ is below $\eta(\frac{1}{k} - \frac{n}{k|\mathbb{F}|})$, then *every* list of $\text{RS}[\mathbb{F}, D, k + 1]$ at radius δ has size at most $\lceil |\mathbb{F}| \varepsilon_{\text{ca}} / (1 - \eta) \rceil$. Taking $\eta = \frac{1}{2}$, the heavy fiber of Lemma 3.1 violates this whenever $\binom{N}{\ell} \geq |B|(|\mathbb{F}|/k + 1)$ — an inequality with hundreds of bits of slack throughout the challenge envelope — and the contrapositive yields $\varepsilon_{\text{ca}}(C, \delta) > \frac{1}{2k}(1 - n/|\mathbb{F}|)$ throughout the integer-admissible range. The MCA cap then persists up to capacity by support-wise MCA monotonicity.

No value-set counting, exponential-sum estimate, or equidistribution hypothesis enters: pigeonhole fibers exist over every field unconditionally, and Theorem 2.6 converts list mass into correlated-agreement error directly. The cost of this route, relative to the error-one results of [Cho26a, Cho26b] below 2^{150} , is an error of order $1/k$ rather than $1 - o(1)$, and a gap 2^{-9} at the first three prize rates (uniformly 2^{-10} across all four), rather than the fixed gap $1/64$; Section 8 quantifies the trade-off and what remains open.

Beyond the universal cap (Corollary 5.1, with a two-tier summary combining it with [Cho26b]), we prove: a deployed-parameter corollary at the KoalaBear-sextic instantiation of [ABF26, §6.3], with $\varepsilon_{\text{mca}} > 2^{-22}$ at gap 2^{-7} and the same ε_{ca} lower bound on the integer-admissible subinterval (Corollary 5.4), extended to every interleaved row of [ABF26, Tables 2–3] via an interleaving-transfer lemma (Lemma 3.68 and Corollary 5.5); an independent slacked variant through [BCHKS25, Thm. 1.9] for robustness (Proposition 5.6); and a subfield-confinement lemma (Lemma 6.1) showing that for lines with B -valued words over an extension $\mathbb{F}/B$, every CA- or MCA-bad slope lies in B . The latter has a structural consequence (Corollary 6.2): at deployed parameters the lines certifying Corollary 5.4 are necessarily *not* B -rational, so the failure proved here is fiber-borne and currently non-constructive; exhibiting explicit certifying lines is posed as Open Problem 8.1.

Roadmap. Section 2 fixes definitions, aligned with [ABF26, Defs. 4.1, 4.3], proves the direct deep-point conversion used for the main cap, and records the separate BCHKS/ABF slacked fallback. Section 3 proves the fiber and interleaving lemmas and the conservative quotient-support upper ledger, Section 4 the main theorem, Section 5 the challenge-level and deployed corollaries, and Section 6 subfield confinement. The new Section 7 states exactly what this paper contributes toward the Proximity Prize [PP26], gives conditional completion theorems for the MCA and interleaved-list thresholds, and isolates the proof obligations that would turn the present cap into a full threshold determination. Section 8 discusses scope, limits, calibration against the conjectural picture of [Cho26b], and open problems. Throughout, companion results of [Cho26a, Cho26b] are cited by their printed theorem names rather than numbers, since the numbering of those manuscripts is not yet frozen.

2 Preliminaries

Codes and distances. Throughout, $B \subseteq \mathbb{F}$ are finite fields, $q := |\mathbb{F}|$, and $D = \alpha H \subseteq B^\times$ is a multiplicative coset of a cyclic subgroup H of order n ; we write D for the evaluation domain (denoted L in [ABF26]). The subgroup case is $\alpha = 1$, and taking $B = \mathbb{F}$ is always allowed. For

$k \leq n$,

$$\text{RS}[\mathbb{F}, D, k] := \{(p(x))_{x \in D} : p \in \mathbb{F}[X], \deg p < k\} \subseteq \mathbb{F}^n, \quad \rho := k/n.$$

For $u, v \in \mathbb{F}^n$, $\Delta(u, v)$ is the relative Hamming distance, $\Delta(u, C) := \min_{c \in C} \Delta(u, c)$, and the relative minimum distance of $C = \text{RS}[\mathbb{F}, D, k]$ is $\delta_{\min}(C) = 1 - \rho + 1/n$. For $S \subseteq D$, $\Delta_S(u, v)$ and $\Delta_S(u, C)$ denote the corresponding restricted relative distances on S . Lists are denoted

$$\Lambda(C, \delta, u) := \{c \in C : \Delta(u, c) \leq \delta\}, \quad \Lambda(C, \delta) := \max_{u \in \mathbb{F}^n} |\Lambda(C, \delta, u)|.$$

For $s \geq 1$ the s -interleaved code is $C^{\equiv s} := \{(c^{(1)}, \dots, c^{(s)}) : c^{(i)} \in C\} \subseteq (\mathbb{F}^n)^s$ with the column distance $\Delta_s(F, G) := \frac{1}{n} |\{j : \exists i, f_j^{(i)} \neq g_j^{(i)}\}|$, and $\Delta_s(F, C^{\equiv s}) := \min_{\mathbf{c} \in C^{\equiv s}} \Delta_s(F, \mathbf{c})$.

Definition 2.1 (correlated agreement error; [ABF26, Def. 4.1] up to notation). For $0 \leq \delta_{\text{fld}} \leq \delta_{\text{int}} < 1$,

$$\varepsilon_{\text{ca}}(C, \delta_{\text{fld}}, \delta_{\text{int}}) := \max_{f_1, f_2 \in \mathbb{F}^n} \Pr_{\gamma \leftarrow \mathbb{F}} \left[\Delta(f_1 + \gamma f_2, C) \leq \delta_{\text{fld}} \wedge \Delta_2((f_1, f_2), C^{\equiv 2}) > \delta_{\text{int}} \right],$$

and the no-proximity-loss version is $\varepsilon_{\text{ca}}(C, \delta) := \varepsilon_{\text{ca}}(C, \delta, \delta)$. A slope γ realizing the event for a fixed pair (f_1, f_2) is called *CA-bad* for that pair.

Definition 2.2 (mutual correlated agreement error, support-wise; [ABF26, Def. 4.3] up to notation). For $\delta \in (0, 1)$,

$$\varepsilon_{\text{mca}}(C, \delta) := \max_{f_1, f_2 \in \mathbb{F}^n} \Pr_{\gamma \leftarrow \mathbb{F}} \left[\begin{array}{l} \exists S \subseteq D, |S| \geq (1 - \delta)n, \Delta_S(f_1 + \gamma f_2, C) = 0, \\ \text{and } \Delta_S((f_1, f_2), C^{\equiv 2}) > 0 \end{array} \right].$$

Equivalently, γ is MCA-bad for (f_1, f_2) if some large agreement set S explains the point $f_1 + \gamma f_2$, but the same set S does not simultaneously explain f_1 and f_2 by two codewords of C .

Definition 2.3 (challenge threshold). Following the grand MCA challenge of [ABF26, §1], for $\varepsilon^* \in (0, 1)$ set $\delta_C^*(\varepsilon^*) := \sup\{\delta \in (0, 1 - \rho) : \varepsilon_{\text{mca}}(C, \delta) \leq \varepsilon^*\}$, with $\sup \emptyset := 0$.

Fact 2.4 (cf. [ABF26, Fact 4.5]). For every $\delta \in (0, 1)$: $\varepsilon_{\text{ca}}(C, \delta) \leq \varepsilon_{\text{mca}}(C, \delta)$.

Proof. Fix (f_1, f_2) and let γ be CA-bad: $\Delta(f_1 + \gamma f_2, C) \leq \delta$ and $\Delta_2((f_1, f_2), C^{\equiv 2}) > \delta$. Choose $S \subseteq D$ of size at least $(1 - \delta)n$ on which $f_1 + \gamma f_2$ agrees with a codeword of C . Since (f_1, f_2) is not δ -close to $C^{\equiv 2}$, there is no set of size at least $(1 - \delta)n$ on which f_1 and f_2 are simultaneously explained by codewords of C ; in particular this fails on the chosen S . Thus γ is MCA-bad for the same pair. Taking maxima over pairs gives the claim. $\square$

Lemma 2.5 (support-wise MCA monotonicity). For every linear code $C \subseteq \mathbb{F}^D$, the function

$$\delta \mapsto \varepsilon_{\text{mca}}(C, \delta)$$

is nondecreasing.

Proof. Fix a pair (f_1, f_2) and suppose that γ is MCA-bad at radius δ . Then there is a set $S \subseteq D$ with

$$|S| \geq (1 - \delta)n$$

such that $f_1 + \gamma f_2$ is explained by C on S , but (f_1, f_2) is not simultaneously explained by two codewords of C on the same S . If $\delta' \geq \delta$, then

$$|S| \geq (1 - \delta)n \geq (1 - \delta')n,$$

so the same set S witnesses that γ is MCA-bad at radius δ' . Taking the maximum over pairs gives the claim. $\square$

Conversion theorem. The main composition below uses the deep-point list-to-CA conversion proved here directly. This is the same consequence previously imported from Crites–Stewart, with the integer-radius admissibility condition made explicit; see Remark 2.17. We keep the slacked BCHKS/ABF fallback as a separate imported route.

Theorem 2.6 (deep-point list-to-CA conversion). *Let $C = \text{RS}[\mathbb{F}, D, k]$, let $C^+ := \text{RS}[\mathbb{F}, D, k + 1]$, let $q := |\mathbb{F}|$ and $n := |D|$, and assume $q > n$. Let $\delta \in (0, 1)$ and set $f_\delta := \lfloor \delta n \rfloor$. Assume*

$$f_\delta \leq n - k - 1.$$

For any $\eta \in [0, 1)$, if

$$\varepsilon_{\text{ca}}(C, \delta) \leq \eta \left(\frac{1}{k} - \frac{n}{kq} \right),$$

then

$$\Lambda(C^+, \delta) \leq \left\lceil \frac{q \varepsilon_{\text{ca}}(C, \delta)}{1 - \eta} \right\rceil.$$

Proof. Put $\epsilon := \varepsilon_{\text{ca}}(C, \delta)$ and $f = f_\delta$. Let $U : D \rightarrow \mathbb{F}$ be any received word and suppose that $P_1, \dots, P_L \in \mathbb{F}[X]_{<k+1}$ are distinct polynomials whose restrictions to D all lie in the Hamming ball of relative radius δ around U . Since distances are integral, each P_i agrees with U on at least

$$a := n - f$$

points of D . The hypothesis $f \leq n - k - 1$ gives $a \geq k + 1$, in particular $a > k$.

Let $\Omega := \mathbb{F} \setminus D$, so $|\Omega| = q - n$. For each $\alpha \in \Omega$ define the simple-pole line

$$f_\alpha(x) = \frac{U(x)}{x - \alpha}, \quad g_\alpha(x) = -\frac{1}{x - \alpha} \quad (x \in D).$$

If P_i agrees with U on a set S_i of size at least a , then for $z_i(\alpha) := P_i(\alpha)$ and every $x \in S_i$,

$$f_\alpha(x) + z_i(\alpha)g_\alpha(x) = \frac{P_i(x) - P_i(\alpha)}{x - \alpha}.$$

The right-hand side is the restriction of a polynomial of degree $< k$. Thus each distinct value among the $P_i(\alpha)$ gives a slope for which the line point is δ -close to C .

We next check the correlated-agreement far condition. If a polynomial $G \in \mathbb{F}[X]_{<k}$ agreed with g_α on more than k points of D , then

$$(X - \alpha)G(X) + 1$$

would be a polynomial of degree at most k with more than k roots, but its value at $X = \alpha$ is 1, a contradiction. Hence (f_α, g_α) is not δ -close to $C^{\equiv 2}$, because any common explanation on a support of size at least $a > k$ would in particular explain g_α there. Therefore the distinct values of $P_i(\alpha)$ are CA-bad slopes.

It remains to lower-bound the number of distinct values for some α . For $i \neq j$, the polynomial $P_i - P_j$ is nonzero of degree at most k , so it has at most k roots in Ω . Let $M(\alpha)$ be the number of distinct values among $P_1(\alpha), \dots, P_L(\alpha)$. Over all $\alpha \in \Omega$, the total number of colliding unordered pairs (i, j) with $P_i(\alpha) = P_j(\alpha)$ is at most $k \binom{L}{2}$. Equivalently, if the fiber sizes of the map $i \mapsto P_i(\alpha)$ are $m_v(\alpha)$, then

$$\sum_{\alpha \in \Omega} \sum_v \binom{m_v(\alpha)}{2} \leq k \binom{L}{2}.$$

Choose $\alpha \in \Omega$ for which the inner collision count is at most $k\binom{L}{2}/(q-n)$. If the fiber sizes of the map $i \mapsto P_i(\alpha)$ are $m_1, \dots, m_{M(\alpha)}$, then

$$\sum_r m_r = L, \quad \sum_r m_r^2 = L + 2 \sum_r \binom{m_r}{2} \leq L + \frac{kL(L-1)}{q-n}.$$

Cauchy–Schwarz therefore gives

$$L^2 \leq M(\alpha) \sum_r m_r^2, \quad M(\alpha) \geq \frac{L(q-n)}{q-n+k(L-1)} \geq \frac{L(q-n)}{q-n+kL}.$$

Consequently

$$\epsilon \geq \frac{1}{q} \cdot \frac{L(q-n)}{q-n+kL}.$$

Solving this inequality for L gives

$$L \leq \frac{\epsilon q(q-n)}{q-n-k\epsilon q},$$

provided $\epsilon < (q-n)/(kq)$. Under the stated hypothesis, $\epsilon \leq \eta(q-n)/(kq)$ with $\eta < 1$, so the denominator is at least $(1-\eta)(q-n)$ and hence

$$L \leq \frac{q\epsilon}{1-\eta}.$$

Since U and the list were arbitrary, the displayed ceiling bounds the maximum list size of C^+ at radius δ . $\square$

Remark 2.7 (relation to Crites–Stewart). The displayed conversion is the only consequence of Crites–Stewart needed by the universal-cap argument. Earlier drafts imported it from [CS25, Thm. 2]; the proof above is included to make the main cap self-contained. The separate slacked fallback through BCHKS/ABF remains an independent imported route.

Proposition 2.8 (quantitative deep-list floor). *Let $C = \text{RS}[\mathbb{F}, D, k]$, let $C^+ := \text{RS}[\mathbb{F}, D, k+1]$, put $q = |\mathbb{F}|$ and $n = |D|$, and assume $q > n$. Let $\delta_0 \in [0, 1 - k/n)$, and suppose that for some received word $U : D \rightarrow \mathbb{F}$ there are $L \geq 1$ distinct codewords of C^+ in the closed ball of radius δ_0 around U . Then, for every $\delta \in [\delta_0, 1 - k/n)$,*

$$\varepsilon_{\text{ca}}(C, \delta) \geq \frac{L(q-n)}{q(q-n+kL)}.$$

For every such $\delta > 0$,

$$\varepsilon_{\text{mca}}(C, \delta) \geq \frac{L(q-n)}{q(q-n+kL)}$$

as well. More precisely, some received line has at least

$$\left\lceil \frac{L(q-n)}{q-n+kL} \right\rceil$$

CA-bad finite slopes at every such radius; for $\delta > 0$, the same slopes are MCA-bad.

Proof. Fix $\delta \in [\delta_0, 1 - k/n)$ and choose distinct $P_1, \dots, P_L \in \mathbb{F}[X]_{<k+1}$ in the δ_0 -ball around U . They are also in the δ -ball. Since $\delta < 1 - k/n$, if $f = \lfloor \delta n \rfloor$ then $f \leq n - k - 1$, so every codeword in this δ -ball agrees with U on at least $a = n - f > k$ points.

Repeat the simple-pole construction in the proof of Theorem 2.6. For $\alpha \in \Omega := \mathbb{F} \setminus D$ set

$$f_\alpha(x) = \frac{U(x)}{x - \alpha}, \quad g_\alpha(x) = -\frac{1}{x - \alpha} \quad (x \in D).$$

For every distinct value of $P_i(\alpha)$, the word $f_\alpha + P_i(\alpha)g_\alpha$ is δ -close to C . The pair (f_α, g_α) is not δ -close to $C^{\equiv 2}$, because a common explanation on more than k points would in particular explain g_α there, and then $(X - \alpha)G(X) + 1$ would be a polynomial of degree at most k with more than k roots and value 1 at α . Thus the distinct values of the map $i \mapsto P_i(\alpha)$ are CA-bad slopes.

For $i \neq j$, the polynomial $P_i - P_j$ has degree at most k and is nonzero, hence it has at most k roots in Ω . Averaging over $|\Omega| = q - n$ gives an α for which the number of colliding unordered pairs among the L values is at most $k \binom{L}{2} / (q - n)$. If $m_1, \dots, m_M$ are the value multiplicities for this α , then

$$\sum_r m_r = L, \quad \sum_r m_r^2 \leq L + \frac{kL(L-1)}{q-n}.$$

By Cauchy–Schwarz,

$$M \geq \frac{L^2}{L + kL(L-1)/(q-n)} = \frac{L(q-n)}{q-n + k(L-1)} \geq \frac{L(q-n)}{q-n + kL}.$$

Since M is an integer, $M \geq \lceil L(q-n)/(q-n + kL) \rceil$. Dividing by the q possible finite slopes gives the CA lower bound. For $\delta > 0$, the MCA bound follows from Fact 2.4. $\square$

Corollary 2.9 (deep-list trigger and ceiling). *In the setting of Proposition 2.8, define*

$$\mathcal{E}_{q,k}(L) := \frac{L(q-n)}{q(q-n + kL)}, \quad \mathcal{N}_{q,k}(L) := \left\lceil \frac{L(q-n)}{q-n + kL} \right\rceil.$$

Then $\mathcal{E}_{q,k}(L)$ is increasing in L , and

$$\mathcal{E}_{q,k}(L) > \frac{1}{2k} \left(1 - \frac{n}{q}\right) \iff L > \frac{q-n}{k}.$$

Moreover

$$\mathcal{E}_{q,k}(L) < \frac{1}{k} \left(1 - \frac{n}{q}\right)$$

for every finite L , and the right-hand side is the limiting value as $L \rightarrow \infty$.

Proof. The derivative of $L/(q-n + kL)$ is positive. The trigger inequality is

$$\frac{L}{q-n + kL} > \frac{1}{2k},$$

which is equivalent to $kL > q - n$. The upper bound follows from $q - n + kL > kL$. $\square$

Remark 2.10 (why this is stronger than the contrapositive). Theorem 2.6 is optimized for contradiction: once $L > (q - n)/k$, taking $\eta = 1/2$ already forces the printed $(2k)^{-1}(1 - n/q)$ lower bound. Proposition 2.8 keeps the whole interpolation count. It is the form a staircase scanner should use when a quotient fiber is too small to cross the $1/(2k)$ trigger but still contributes a nonzero explicit bad-slope numerator, and also when a very large fiber improves the constant toward $(1/k)(1 - n/q)$.

Proposition 2.11 (punctured simple-pole deep-list floor). *Let $C = \text{RS}[\mathbb{F}, D, k]$, let $C^+ := \text{RS}[\mathbb{F}, D, k + 1]$, put $q = |\mathbb{F}|$ and $n = |D|$. Let*

$$\Omega_0 \subseteq \mathbb{F} \setminus D, \quad m := |\Omega_0| > 0.$$

Let $\delta_0 \in [0, 1 - k/n)$, and suppose that for some received word $U : D \rightarrow \mathbb{F}$ there are $L \geq 1$ distinct codewords of C^+ in the closed ball of radius δ_0 around U . Then, for every $\delta \in [\delta_0, 1 - k/n)$,

$$\varepsilon_{\text{ca}}(C, \delta) \geq \frac{Lm}{q(m + kL)}.$$

For every such $\delta > 0$,

$$\varepsilon_{\text{mca}}(C, \delta) \geq \frac{Lm}{q(m + kL)}.$$

More precisely, there is some $\alpha \in \Omega_0$ such that the simple-pole line

$$f_\alpha(x) = \frac{U(x)}{x - \alpha}, \quad g_\alpha(x) = -\frac{1}{x - \alpha} \quad (x \in D)$$

has at least

$$\left\lfloor \frac{Lm}{m + kL} \right\rfloor$$

CA-bad finite slopes at every such radius; for $\delta > 0$, the same slopes are MCA-bad.

Proof. Repeat the proof of Proposition 2.8, but average only over Ω_0 instead of over all of $\mathbb{F} \setminus D$. The simple-pole argument works for every $\alpha \notin D$, so every distinct value among $P_1(\alpha), \dots, P_L(\alpha)$ gives a CA-bad slope. For $i \neq j$, the nonzero polynomial $P_i - P_j$ has degree at most k , hence has at most k roots inside the smaller set Ω_0 . Therefore the total number of colliding unordered pairs over all $\alpha \in \Omega_0$ is at most $k \binom{L}{2}$. Choose $\alpha \in \Omega_0$ for which the collision count is at most $k \binom{L}{2} / m$. If M is the number of distinct values at this α , the same Cauchy–Schwarz calculation gives

$$M \geq \frac{Lm}{m + k(L - 1)} \geq \frac{Lm}{m + kL}.$$

Since M is integral, this gives the displayed bad-slope numerator. Dividing by the q finite slopes gives the CA lower bound, and the MCA statement for $\delta > 0$ follows from Fact 2.4. $\square$

Corollary 2.12 (extension-pole deep-list floor). *Assume in addition that $B \subsetneq \mathbb{F}$, $D \subseteq B$, and $|D| \geq 2$. Put $b := |B|$. In the setting of Proposition 2.11, take*

$$\Omega_0 := \mathbb{F} \setminus B, \quad m = q - b.$$

Then for every $\delta \in [\delta_0, 1 - k/n)$,

$$\varepsilon_{\text{ca}}(C, \delta) \geq \mathcal{E}_{q,b,k}^{\text{ext}}(L) := \frac{L(q - b)}{q(q - b + kL)}.$$

For $\delta > 0$ the same bound holds for $\varepsilon_{\text{mca}}(C, \delta)$. Moreover the witnessing line may be chosen with pole $\alpha \in \mathbb{F} \setminus B$, and its pole vector $((x - \alpha)^{-1})_{x \in D}$ is not a scalar multiple over $\mathbb{F}$ of any B -valued vector. In particular the witness line is genuinely extension-valued, not merely a B -rational line viewed over $\mathbb{F}$.

Proof. The lower bound is Proposition 2.11 with $\Omega_0 = \mathbb{F} \setminus B$. It remains to prove genuine extension-valuedness. Suppose that, for some $\lambda \in \mathbb{F}^\times$, all numbers $\lambda/(x - \alpha)$ with $x \in D$ lay in B . Choose two distinct points $x, y \in D$. Then

$$\frac{y - \alpha}{x - \alpha} = \frac{\lambda/(x - \alpha)}{\lambda/(y - \alpha)} \in B.$$

Writing this ratio as $r \in B$, we cannot have $r = 1$ because $x \neq y$. Hence

$$\alpha = \frac{rx - y}{r - 1} \in B,$$

contradicting $\alpha \notin B$. Thus the pole vector is not projectively B -valued. In particular the displayed affine parametrization has no B -valued direction vector after rescaling. $\square$

Corollary 2.13 (extension-pole quotient-remainder floor). *Assume $B \subsetneq \mathbb{F}$, $D \subseteq B^\times$ is a multiplicative coset of order $n \geq 2$, put $b := |B|$, and let $C = \text{RS}[\mathbb{F}, D, k]$. Fix a quotient-remainder profile as in Lemma 3.7 with $K = k + 1$ and agreement $A_0 > k$. Let*

$$L := H_{c,m,s}^{k+1}$$

be the certified heaviest prefix-fiber size, or replace L by either fallback

$$\left\lceil \frac{M_{c,m,s}}{I_{c,m,s}^{k+1}} \right\rceil, \quad \left\lceil \frac{M_{c,m,s}}{|B|^{w_c(s, A_0 - k - 1)}} \right\rceil.$$

Then, for every

$$\delta \in [1 - A_0/n, 1 - k/n),$$

there is a genuinely extension-valued simple-pole line with at least

$$\mathcal{N}_{q,b,k}^{\text{ext}}(L) := \left\lceil \frac{L(q - b)}{q - b + kL} \right\rceil$$

CA-bad finite slopes. For $\delta > 0$, these slopes are MCA-bad. Equivalently,

$$\varepsilon_{\text{ca}}(C, \delta) \geq \frac{L(q - b)}{q(q - b + kL)}, \quad \varepsilon_{\text{mca}}(C, \delta) \geq \frac{L(q - b)}{q(q - b + kL)} \quad (\delta > 0).$$

If $L > (q - b)/k$, this gives the clean trigger

$$\varepsilon_{\text{ca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{b}{q}\right), \quad \varepsilon_{\text{mca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{b}{q}\right) \quad (\delta > 0).$$

Proof. Lemma 3.7 gives a B -valued received word with at least L codewords of C^+ in the closed ball of radius $1 - A_0/n$. Apply Corollary 2.12. The trigger inequality is

$$\frac{L(q - b)}{q(q - b + kL)} > \frac{q - b}{2kq},$$

which is equivalent to $kL > q - b$. $\square$

Definition 2.14 (extension-pole certificate). An extension-pole certificate for a list $P_1, \dots, P_L \in \mathbb{F}[X]_{<k+1}$ and a received word U consists of:

- (i) a declared proper subfield $B \subsetneq \mathbb{F}$ containing D and a pole $\alpha \in \mathbb{F} \setminus B$;
- (ii) the agreement supports showing that each P_i lies in the declared ball around U ;
- (iii) the bucket table for the values $P_i(\alpha)$, with duplicate values coalesced;
- (iv) the simple-pole line (f_α, g_α) and the proof that g_α is not projectively B -valued;
- (v) the final bad-slope numerator, namely the number of distinct buckets in (iii).

The verifier checks $\alpha \notin B$, the support agreements, the value buckets, and the simple-pole far condition. For quotient-remainder floors, the list in (ii) may itself be supplied by a prefix-fiber certificate from Definition 3.6.

Remark 2.15 (what the extension-pole milestone changes). The subfield-confinement theorem says that B -rational lines are inert over a proper extension. Corollary 2.12 gives the matching existence theorem on the failure side: any large list can be converted using poles restricted to $\mathbb{F} \setminus B$, so the resulting bad line is genuinely extension-valued. Thus the extension-line bucket in a prize scanner is no longer only an abstract warning; it has an explicit simple-pole source and a stable numerator with $q - b$ replacing $q - n$. What remains open is not existence of extension-valued witnesses, but their sharp classification inside the aperiodic/residue-line ledgers and, for a fully explicit counterexample, printing a concrete pole α with its value-bucket table.

Theorem 2.16 ([BCHKS25, Thm. 1.9], as stated in [ABF26, Thm. 5.2]). *Let $C = \text{RS}[\mathbb{F}, D, k]$, let $\rho = k/n$, and let $\delta \in (0, 1 - \rho)$ satisfy $\delta + 2/n \leq 1 - \rho - 1/n$. If*

$$\varepsilon_{\text{ca}}\left(C, \delta + \frac{2}{n}, 1 - \rho - \frac{1}{n}\right) < \frac{1}{2n}, \quad \text{then} \quad \Lambda(C, \delta) < |\mathbb{F}|.$$

Remark 2.17 (on imports and radius conventions). Theorem 2.6 is the only Crites–Stewart-type consequence used by the main cap, and it is now proved above by the simple-pole deep-point argument. The important admissibility condition is the integer-radius condition

$$\lfloor \delta n \rfloor \leq n - k - 1.$$

Consequently, the no-loss CA lower bound proved below should be read only in this admissible range. The Proximity-Prize MCA cap is unaffected: we prove the lower bound at the endpoint

$$\delta_N := 1 - \rho - \frac{2}{N}$$

and then use support-wise MCA monotonicity to extend the MCA failure to all larger sub-capacity radii.

Theorem 2.16 remains a separate slacked fallback imported through the BCHKS/ABF route. It is not needed for the main field-size-universal MCA cap.

3 Locator fibers and interleaving transfer

Fix $N \mid n$, set $a := n/N$, and let $Q := D^a := \{x^a : x \in D\} \subseteq B^\times$, a multiplicative coset of order N . The a -th power map $D \rightarrow Q$ is surjective and its fibers $S_b := \{x \in D : x^a = b\}$ have exactly a elements each. Since $a \mid n \mid |B| - 1$, the characteristic of $\mathbb{F}$ does not divide a , so $X^a - b$ is separable for $b \neq 0$ and its full root set inside D is S_b whenever $b \in Q$. For a set A , $e_j(A)$ denotes the j -th elementary symmetric function of its elements.

Lemma 3.1 (locator fibers are lists). *Let $B \subseteq \mathbb{F}$, let $D \subseteq B^\times$ be a multiplicative coset of order n , and let k satisfy $a \mid k$ and $\rho = k/n$. Write $\ell_1 := \rho N + 1$ and $\ell_2 := \rho N + 2$.*

(i) *If $\ell_1 \leq N$, there exists $z \in B$ such that, with $u_z := (x^{k+a} + z x^k)_{x \in D} \in \mathbb{F}^n$,*

$$|\Lambda(\text{RS}[\mathbb{F}, D, k], 1 - \rho - \frac{1}{N}, u_z)| \geq \binom{N}{\ell_1} / |B|.$$

(ii) *If $\ell_2 \leq N$, there exists $z \in B$ such that, with $u_z := (x^{k+2a} + z x^{k+a})_{x \in D} \in \mathbb{F}^n$,*

$$|\Lambda(\text{RS}[\mathbb{F}, D, k+1], 1 - \rho - \frac{2}{N}, u_z)| \geq \binom{N}{\ell_2} / |B|.$$

In both cases the words u_z are B -valued and all listed codewords lie in $\text{RS}[B, D, k]$ (resp. $\text{RS}[B, D, k+1]$).

Proof. We prove (ii); part (i) is the identical computation one degree rung lower and is [Cho26a, Main Thm. (d)] sharpened by pigeonholing over B in place of $\mathbb{F}$.

Let $A \subseteq Q$ with $|A| = \ell_2$, and put

$$L_A(X) := \prod_{b \in A} (X^a - b) = \sum_{j=0}^{\ell_2} (-1)^j e_j(A) X^{a(\ell_2-j)} = X^{k+2a} - e_1(A) X^{k+a} + R_A(X),$$

where $a\ell_2 = k + 2a$, $a(\ell_2 - 1) = k + a$, and R_A collects the terms with $j \geq 2$, so $\deg R_A \leq a(\ell_2 - 2) = k$. The polynomial L_A is squarefree with root set $S_A := \bigcup_{b \in A} S_b \subseteq D$ of size $a\ell_2 = k + 2a$.

Set $z_A := -e_1(A) \in B$ and $c_A := (-R_A(x))_{x \in D}$. Since $\deg R_A \leq k$, we have $c_A \in \text{RS}[B, D, k+1] \subseteq \text{RS}[\mathbb{F}, D, k+1]$. For every $x \in S_A$,

$$0 = L_A(x) = x^{k+2a} + z_A x^{k+a} + R_A(x), \quad \text{i.e.} \quad u_{z_A}(x) = c_A(x),$$

so u_{z_A} and c_A agree on at least $k + 2a$ points of D and $\Delta(u_{z_A}, c_A) \leq 1 - \rho - 2/N$.

The map $A \mapsto c_A$ is injective on the ℓ_2 -subsets of Q sharing a common slope value: if $z_A = z_{A'}$ and $c_A = c_{A'}$, then R_A and $R_{A'}$ are polynomials of degree $\leq k < n$ agreeing on all n points of D , hence equal as polynomials; then $L_A = L_{A'}$, so $S_A = S_{A'}$, and $A = \{x^a : x \in S_A\} = A'$.

Finally, $z_A = -e_1(A)$ takes at most $|B|$ values as A ranges over the $\binom{N}{\ell_2}$ subsets; some $z \in B$ is attained by at least $\binom{N}{\ell_2} / |B|$ of them, and the corresponding codewords c_A are pairwise distinct elements of $\Lambda(\text{RS}[\mathbb{F}, D, k+1], 1 - \rho - 2/N, u_z)$. $\square$

Remark 3.2 (why slack two). Part (i) applied to $C^+ = \text{RS}[\mathbb{F}, D, k+1]$ directly would require $a \mid k+1$; in every challenge instantiation a is a power of two and $k+1$ is odd, so this fails. At slack two the listed codewords have degree $\leq k$, i.e. lie in C^+ , while the divisibility demand stays on k . This is exactly the interface needed by Theorem 2.6, whose list conclusion concerns C^+ while its hypothesis concerns C .

Remark 3.3 (the subfield pigeonhole elsewhere). The same one-line strengthening — pigeonholing locator data over the field of definition B rather than over $\mathbb{F}$ — also upgrades the coefficient-pigeonhole bound of [Cho26b] to the generated field: the prefix map Φ_σ there has image in B^σ whenever $D \subseteq B$, since the elementary symmetric prefixes of subsets of D are B -valued. This is the form consumed by the “necessary” half of the list-profile conjecture of [Cho26b], which states the entropy floor at the generated field $\tau^*(\rho, q_D)$; Lemma 3.1 supplies the missing subfield step.

Remark 3.4 (lists, not lines). The words u_z of Lemma 3.1 lie on the B -rational line $f + zg$ with $f = (x^{k+ta})_{x \in D}$, $g = (x^{k+(t-1)a})_{x \in D}$, but the lemma is purely a statement about lists at a received word and no line structure is used in Section 4. This matters: by Corollary 6.2 below, over a proper extension $\mathbb{F} \supsetneq B$ the lines that end up certifying the CA failure cannot be these (or any) B -rational lines.

3.1 Quotient-profile floors with remainder supports

The fiber lemma above uses supports that are unions of complete quotient fibers. For threshold work one also needs agreements that are not multiples of the fiber size. The following locator-prefix pigeonhole gives a finite lower ledger for such remainder supports. It is a lower-floor theorem only; the matching upper ledger over all divisor scales remains part of the prize-completion program.

Let $c \mid n$ and put $Q_c := D^c$, so the map

$$\pi_c : D \rightarrow Q_c, \quad x \mapsto x^c$$

has fibers of size c and $|Q_c| = N = n/c$. For $0 \leq s < c$ and an integer $\sigma \geq 0$, define the prefix weight

$$w_c(s, \sigma) := \#\{h : 1 \leq h \leq \sigma, h \bmod c \in \{0, 1, \dots, s\}\},$$

where residues are taken in $\{0, 1, \dots, c-1\}$. Thus $w_c(0, \sigma) = \lfloor \sigma/c \rfloor$ for full-fiber supports.

Lemma 3.5 (quotient-remainder locator-prefix lower floor). *Let $B \subseteq \mathbb{F}$, let $D \subseteq B^\times$ be a multiplicative coset of order n , and let $K < n$. Fix $c \mid n$, write $N = n/c$, and let*

$$A_0 = mc + s, \quad 0 \leq s < c, \quad 0 \leq m \leq N, \quad A_0 \geq K.$$

If $s > 0$, assume $mc + s \leq n$. Put $\sigma = A_0 - K$ and

$$M_{c,m,s} := \binom{N}{m} \binom{n-mc}{s}.$$

Then there is a B -valued received word $U : D \rightarrow B$ such that

$$|\Lambda(\text{RS}[\mathbb{F}, D, K], 1 - A_0/n, U)| \geq \left\lceil \frac{M_{c,m,s}}{|B|^{w_c(s, \sigma)}} \right\rceil.$$

The listed codewords may be chosen in $\text{RS}[B, D, K]$.

Proof. For $M \subseteq Q_c$ with $|M| = m$, write

$$P_M(X) = \prod_{b \in M} (X^c - b).$$

For $R \subseteq D \setminus \pi_c^{-1}(M)$ with $|R| = s$, put

$$E_R(X) = \prod_{r \in R} (X - r), \quad L_{M,R}(X) = P_M(X)E_R(X).$$

Then $L_{M,R}$ is a squarefree locator of degree A_0 whose roots are exactly $\pi_c^{-1}(M) \cup R$. Write

$$L_{M,R}(X) = X^{A_0} + \sum_{h=1}^{A_0} \lambda_h(M, R) X^{A_0-h}.$$

The coefficients lie in B . Moreover, because P_M has monomials only in degrees congruent to 0 modulo c and E_R has degree s , the coefficient $\lambda_h(M, R)$ is zero unless $h \bmod c \in \{0, 1, \dots, s\}$. Consequently the high-degree prefix

$$(\lambda_h(M, R) : 1 \leq h \leq \sigma, \ h \bmod c \in \{0, 1, \dots, s\})$$

takes at most $|B|^{w_c(s, \sigma)}$ values.

Pigeonhole over these prefix values. There is a fixed prefix vector ξ and a subfamily $\mathcal{T}_\xi$ of at least $M_{c,m,s}/|B|^{w_c(s, \sigma)}$ pairs (M, R) with this prefix. Let

$$U_\xi(X) = X^{A_0} + \sum_{1 \leq h \leq \sigma} \lambda_h(\xi) X^{A_0-h}$$

be the corresponding high-degree part, restricted to D . For every $(M, R) \in \mathcal{T}_\xi$, the difference

$$R_{M,R}(X) := L_{M,R}(X) - U_\xi(X)$$

has degree $< K$. Hence the codeword $-R_{M,R}$ agrees with U_ξ on all roots of $L_{M,R}$, a set of size A_0 .

It remains only to check that the listed codewords are distinct. If two pairs (M, R) and (M', R') in the same prefix class gave the same codeword on D , then the two remainders of degree $< K < n$ would be equal as polynomials. Since the high prefixes are also equal, $L_{M,R} = L_{M',R'}$. Their root sets in D are therefore equal. A full fiber has size c while $|R| = s < c$, so the full fibers contained in this root set are exactly M , and then the residual set is forced as well. Thus $(M, R) = (M', R')$. $\square$

Definition 3.6 (prefix fibers and prefix-image certificates). In the notation of Lemma 3.5, let $\Phi_{c,m,s}^K$ be the map from pairs (M, R) to the high-degree prefix

$$(\lambda_h(M, R) : 1 \leq h \leq A_0 - K, \ h \bmod c \in \{0, 1, \dots, s\}) \in B^{w_c(s, A_0-K)}.$$

If $A_0 = K$ this is the unique empty prefix. For a prefix value ξ , put

$$\mu_{c,m,s}^K(\xi) := |(\Phi_{c,m,s}^K)^{-1}(\xi)|.$$

Define the image size and the heaviest prefix fiber by

$$I_{c,m,s}^K := |\text{im } \Phi_{c,m,s}^K|, \quad H_{c,m,s}^K := \max_{\xi} \mu_{c,m,s}^K(\xi).$$

Thus

$$H_{c,m,s}^K \geq \left\lceil \frac{M_{c,m,s}}{I_{c,m,s}^K} \right\rceil \geq \left\lceil \frac{M_{c,m,s}}{|B|^{w_c(s, A_0 - K)}} \right\rceil.$$

A prefix-fiber certificate is an orbit decomposition, hash table, or other verifiable bucket table for the finite map $\Phi_{c,m,s}^K$, including the image size and the maximum fiber size. If only $I_{c,m,s}^K$ is certified, the middle lower bound is still valid; if no image certificate is supplied, the ambient-box lower bound is valid.

Lemma 3.7 (heaviest-prefix locator floor). *Under the hypotheses of Lemma 3.5, there is a B -valued received word $U : D \rightarrow B$ such that*

$$|\Lambda(\text{RS}[\mathbb{F}, D, K], 1 - A_0/n, U)| \geq H_{c,m,s}^K.$$

Consequently

$$|\Lambda(\text{RS}[\mathbb{F}, D, K], 1 - A_0/n, U)| \geq \left\lceil \frac{M_{c,m,s}}{I_{c,m,s}^K} \right\rceil \geq \left\lceil \frac{M_{c,m,s}}{|B|^{w_c(s, A_0 - K)}} \right\rceil$$

when only the coarser certificates are available. The listed codewords may be chosen in $\text{RS}[B, D, K]$.

Proof. Choose a prefix value ξ with $\mu_{c,m,s}^K(\xi) = H_{c,m,s}^K$ and form the high-degree word

$$U_{\xi}(X) = X^{A_0} + \sum_{1 \leq h \leq A_0 - K} \lambda_h(\xi) X^{A_0 - h},$$

where $\lambda_h(\xi) = 0$ for the high coefficients outside the allowed residue classes. For every pair (M, R) in this heaviest prefix fiber, the difference $L_{M,R}(X) - U_{\xi}(X)$ has degree $< K$, so the negative of this remainder is a codeword of $\text{RS}[B, D, K]$ agreeing with U_{ξ} on the A_0 roots of $L_{M,R}$. If two pairs in the same prefix fiber gave the same codeword on D , then their degree- $< K < n$ remainders would be equal as polynomials; together with the common prefix this gives $L_{M,R} = L_{M',R'}$. Their root sets in D are equal. A full quotient fiber has size c while the residual set has size $s < c$, so the full fibers contained in the root set determine M , and then the residual set determines R . Thus the codewords are pairwise distinct. The two coarser displayed bounds follow from Definition 3.6. $\square$

For $A \in \{k+1, \dots, n\}$ and $c \mid n$, write

$$A = m_c(A)c + s_c(A), \quad 0 \leq s_c(A) < c,$$

and set

$$\begin{aligned} M_c(A) &:= \binom{n/c}{m_c(A)} \binom{n - cm_c(A)}{s_c(A)}, \\ L_{c,\text{heavy}}^+(A) &:= H_{c,m_c(A),s_c(A)}^{k+1}, \\ L_{c,\text{image}}^+(A) &:= \left\lceil \frac{M_c(A)}{I_{c,m_c(A),s_c(A)}^{k+1}} \right\rceil, \\ L_{c,\text{box}}^+(A) &:= \left\lceil \frac{M_c(A)}{|B|^{w_c(s_c(A), A-k-1)}} \right\rceil. \end{aligned}$$

The three levels satisfy

$$L_{c,\text{heavy}}^+(A) \geq L_{c,\text{image}}^+(A) \geq L_{c,\text{box}}^+(A).$$

Here “heavy” uses a full prefix-fiber certificate, “image” uses only the prefix-image size, and “box” is the certificate-free bound already implicit in Lemma 3.5.

Theorem 3.8 (quotient-remainder deep-point lower ledger). *Let $B \subseteq \mathbb{F}$, let $q = |\mathbb{F}| > n$, let $D \subseteq B^\times$ be a multiplicative coset of order n , and let $C = \text{RS}[\mathbb{F}, D, k]$. Fix an agreement value $A \in \{k+1, \dots, n\}$ and a divisor $c \mid n$. For each*

$$\star \in \{\text{heavy}, \text{image}, \text{box}\}$$

and every

$$\delta \in [1 - A/n, 1 - k/n)$$

one has

$$\varepsilon_{\text{ca}}(C, \delta) \geq \mathcal{E}_{q,k}(L_{c,\star}^+(A)).$$

If $\delta > 0$, then also

$$\varepsilon_{\text{mca}}(C, \delta) \geq \mathcal{E}_{q,k}(L_{c,\star}^+(A)).$$

Here $\mathcal{E}_{q,k}$ is as in Corollary 2.9. More precisely, some received line has at least

$$\mathcal{N}_{q,k}(L_{c,\star}^+(A))$$

CA-bad finite slopes at that radius, and for $\delta > 0$ these slopes are MCA-bad.

Consequently, for a closed agreement threshold $a \geq k+1$, define

$$B_{Q,\star}^-(a) := \max_{\substack{c \mid n \\ A \in \{a, a+1, \dots, n\}}} \mathcal{N}_{q,k}(L_{c,\star}^+(A)).$$

Then $B_{Q,\star}^-(a)$ is a lower numerator for $q\varepsilon_{\text{ca}}(C, \delta)$ for every $\delta \in [1 - a/n, 1 - k/n)$, and for $q\varepsilon_{\text{mca}}(C, \delta)$ on the same interval with $\delta > 0$. The strongest certified ledger uses $\star = \text{heavy}$; the certificate-free ledger uses $\star = \text{box}$.

Proof. Apply Lemma 3.7 with $K = k+1$ and $A_0 = A$. This produces a $C^+ = \text{RS}[\mathbb{F}, D, k+1]$ list of size at least $L_{c,\text{heavy}}^+(A)$ at radius $1 - A/n$, with the image and ambient-box variants giving the corresponding smaller certified list sizes. Then apply Proposition 2.8. If $A \geq a$ and $\delta \geq 1 - a/n$, then $\delta \geq 1 - A/n$, so the same converted floor applies. The maximum over (c, A) is the correct lower-ledger operation: the definitions of ε_{ca} and ε_{mca} maximize over received lines, so lower witnesses from different divisor scales or exact agreement values need not share one line and must not be summed. $\square$

Corollary 3.9 (quotient-remainder trigger). *In the setting of Theorem 3.8, if $L_{c,\star}^+(A) > (q-n)/k$ for any $\star \in \{\text{heavy}, \text{image}, \text{box}\}$, then, for every $\delta \in [1 - A/n, 1 - k/n)$,*

$$\varepsilon_{\text{ca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{n}{q}\right),$$

and for every such $\delta > 0$,

$$\varepsilon_{\text{mca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{n}{q}\right).$$

Proof. This is Corollary 2.9 applied inside Theorem 3.8. $\square$

Corollary 3.10 (quantitative first-grid floor). *Let $C = \text{RS}[\mathbb{F}, D, k]$, let $q = |\mathbb{F}| > n$, and assume only that $D \subseteq \mathbb{F}$ has n distinct points. Then for every $\delta \in [1 - (k+1)/n, 1 - k/n]$,*

$$\varepsilon_{\text{ca}}(C, \delta) \geq \mathcal{E}_{q,k} \left(\binom{n}{k+1} \right),$$

and for every such $\delta > 0$,

$$\varepsilon_{\text{mca}}(C, \delta) \geq \mathcal{E}_{q,k} \left(\binom{n}{k+1} \right).$$

In particular, if $\binom{n}{k+1} > (q-n)/k$, then the printed $(2k)^{-1}(1 - n/q)$ lower bound holds on this entire first grid interval without any multiplicative smoothness assumption.

Proof. Use the ordinary locator identity for all $(k+1)$ -subsets of D ; this is the $c = 1$ argument behind Lemma 3.7, but it does not use multiplicative smoothness. The prefix is empty, so the list size is at least $\binom{n}{k+1}$. Apply Proposition 2.8 and then Corollary 2.9. $\square$

Definition 3.11 (quotient lower-ledger certificate). A quotient-remainder lower-ledger certificate for a row and a threshold a contains, for each divisor scale c and each exact agreement $A \geq a$ that is used in the maximum:

- (i) the decomposition $A = m_c(A)c + s_c(A)$;
- (ii) the support count $M_c(A)$ and prefix weight $w_c(s_c(A), A - k - 1)$;
- (iii) either a full prefix-fiber bucket table certifying $H_{c, m_c(A), s_c(A)}^{k+1}$, or an image-size certificate for $I_{c, m_c(A), s_c(A)}^{k+1}$, or the declaration that the ambient-box denominator is being used;
- (iv) the resulting list floor $L_{c, \star}^+(A)$;
- (v) the converted slope numerator $\mathcal{N}_{q,k}(L_{c, \star}^+(A))$ and error floor $\mathcal{E}_{q,k}(L_{c, \star}^+(A))$.

The verifier checks the locator-prefix formula, distinctness of codewords inside a prefix fiber, the integer arithmetic, and the final maximum over divisor scales and exact agreement values.

Remark 3.12 (status of T1 after the v10 lower-ledger upgrade). The quotient-profile lower side is now scanner-ready at every agreement value $A = mc + s$: the generated-field contribution is the actual heaviest prefix fiber $H_{c, m, s}^{k+1}$ when certified, the exact prefix-image denominator $I_{c, m, s}^{k+1}$ when only image data are certified, and the ambient box $|B|^{w_c(s, A-k-1)}$ when no certificate is supplied. The challenge-field size q enters only through the independent conversion functions $\mathcal{E}_{q,k}$ and $\mathcal{N}_{q,k}$. Across divisor scales the lower-side operation is a maximum, not a union count, because the witness line may change. The remaining T1 upper-side problem is sharper and unchanged in logical status: replace the safe support ledger and the exact declared-branch image ledger by an exhaustive quotient-periodic classification, with duplicate slopes coalesced across all divisor scales.

Corollary 3.13 (augmented slack-one quotient floor). *Let $B \subseteq \mathbb{F}$ and let $D \subseteq B^\times$ be a multiplicative coset of order n . Put $C = \text{RS}[\mathbb{F}, D, k]$ and $C^+ = \text{RS}[\mathbb{F}, D, k+1]$. If $c \mid \gcd(n, k)$, then there is a B -valued received word U_c such that*

$$|\Lambda(C^+, 1 - (k+c)/n, U_c)| \geq \binom{n/c}{k/c+1}.$$

One may take $U_c(x) = x^{k+c}$. In particular, for $c = 1$ this gives the universal first-grid floor

$$|\Lambda(C^+, 1 - (k+1)/n, x^{k+1})| \geq \binom{n}{k+1},$$

with no quotient denominator. The $c = 1$ statement in fact holds for any set $D \subseteq B$ of n distinct points, not only for multiplicative cosets.

Proof. For $c > 1$, apply Lemma 3.5 with $K = k+1$, $s = 0$, and $m = k/c + 1$. Then $A_0 = k+c$ and $\sigma = A_0 - K = c-1$, so $w_c(0, c-1) = 0$. The high-degree prefix is only the monomial X^{k+c} , giving the displayed word. For $c = 1$, the same argument is just the ordinary locator identity for all $(k+1)$ -subsets of D : the high-degree prefix of a monic locator of degree $k+1$ is X^{k+1} , and its remainder has degree at most k . $\square$

Corollary 3.14 (first-grid deep-point cap). *Let $C = \text{RS}[\mathbb{F}, D, k]$, $q = |\mathbb{F}|$, and suppose $q > n$. Let $c \mid \gcd(n, k)$. If $c > 1$, assume $D \subseteq B^\times$ is a multiplicative coset as above; for $c = 1$, no smoothness is needed. If*

$$\binom{n/c}{k/c+1} \geq q/k + 1,$$

then

$$\varepsilon_{\text{ca}}(C, 1 - (k+c)/n) > \frac{1}{2k} \left(1 - \frac{n}{q}\right), \quad \varepsilon_{\text{mca}}(C, 1 - (k+c)/n) > \frac{1}{2k} \left(1 - \frac{n}{q}\right).$$

For $c = 1$ this is the first closed grid point below capacity.

Proof. Corollary 3.13 gives a C^+ -list of size $L \geq q/k + 1$ at agreement $k+c$. The proof of the deep-point conversion Theorem 2.6 applies whenever the agreement is $> k$, so it applies here for every $c \geq 1$. If the displayed CA lower bound failed, then

$$\varepsilon_{\text{ca}}(C, 1 - (k+c)/n) \leq \frac{q-n}{2kq}.$$

Theorem 2.6 with $\eta = 1/2$ would give

$$\Lambda(C^+, 1 - (k+c)/n) \leq \left\lceil \frac{q-n}{k} \right\rceil < q/k + 1,$$

contradicting the list lower bound. Every no-loss CA-bad slope is also support-wise MCA-bad on the same explaining support, giving the MCA inequality. $\square$

Corollary 3.15 (first-grid cap for large challenge-envelope rows). *Let $\rho \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\}$, let $C = \text{RS}[\mathbb{F}, D, k]$ with $|D| = n$, $k = \rho n$, $q = |\mathbb{F}| < 2^{256}$, and $q > n$. Assume*

$$\frac{\rho}{k} \geq \left| \begin{array}{c} \frac{1}{2} \\ 127 \end{array} \right| \frac{1}{78} \frac{1}{58} \frac{1}{47}$$

for the corresponding rate. Then the first closed grid point below capacity is already CA- and MCA-unsafe:

$$\varepsilon_{\text{ca}}(C, 1 - \rho - 1/n) > \frac{1}{2k} \left(1 - \frac{n}{q}\right), \quad \varepsilon_{\text{mca}}(C, 1 - \rho - 1/n) > \frac{1}{2k} \left(1 - \frac{n}{q}\right).$$

If also $k \leq 2^{40}$, then this lower bound is $> 2^{-86}$ and hence is far above the prize target 2^{-128} .

Proof. By Corollary 3.14 with $c = 1$, it is enough to check

$$\binom{n}{k+1} \geq q/k + 1.$$

Since $q < 2^{256}$, the stronger sufficient condition is $k(\binom{n}{k+1} - 1) \geq 2^{256}$. The four printed lower bounds on k are exactly small finite checks of this inequality at the corresponding rate; the left side is increasing thereafter. For the error size, $q \geq n + 1$, $n = k/\rho \leq 16k$, and $k \leq 2^{40}$ give

$$\frac{1}{2k} \left(1 - \frac{n}{q}\right) \geq \frac{1}{2k(n+1)} > 2^{-86}.$$

□

3.2 A conservative quotient-support upper ledger

The remainder lower floor above supplies the unsafe side of the quotient-profile ledger. For a prize threshold one also needs a safe-side accounting rule: if a later argument declares a bad parameter to be quotient-periodic, the declaration must come with a divisor-lattice numerator that is counted with overlaps under control. The next statements give such a conservative upper ledger. They do not prove that every bad line parameter is quotient-periodic; they only say that once a permitted support family has been specified, every ledger term witnessed inside that family is bounded by the size of the family, with a factor d for finite-parameter degree- d power curves.

Let $\mathcal{S}$ be a family of subsets of D , all of size at least a . For a received line $f + zg$, say that a finite slope is *$\mathcal{S}$ -witnessed support-wise bad* if it has a support-wise noncontained explanation on some $S \in \mathcal{S}$. Define the analogous *$\mathcal{S}$ -witnessed no-loss CA slopes* by requiring that $f + zg$ be explained on some $S \in \mathcal{S}$ while (f, g) is not globally explained on any support of size at least a .

Lemma 3.16 (one support pays for at most one line parameter). *Let $C = \text{RS}[\mathbb{F}, D, k]$ and let $a \geq k$. Fix a support $S \subseteq D$ with $|S| \geq a$. For any received line $f + zg$, at most one finite slope $z \in \mathbb{F}$ can be support-wise noncontained on the fixed support S . Likewise, at most one finite slope can be no-loss CA-bad while being explained on this fixed support S . Consequently, for every support family $\mathcal{S}$,*

$$\#\{\mathcal{S}\text{-witnessed support-wise MCA-bad finite slopes}\} \leq |\mathcal{S}|,$$

and the same bound holds for no-loss CA slopes witnessed by $\mathcal{S}$.

Proof. Suppose two distinct finite slopes $z_1 \neq z_2$ are both explained on the same support S , by codewords $p_1, p_2 \in C$:

$$f + z_i g = p_i \quad \text{on } S, \quad i = 1, 2.$$

Then on S ,

$$g = \frac{p_1 - p_2}{z_1 - z_2}, \quad f = p_1 - z_1 g.$$

Both right-hand sides are codewords of C . Thus (f, g) is simultaneously explained on S . This contradicts support-wise noncontainment on S . It also contradicts no-loss CA badness at agreement a , because $|S| \geq a$ gives a global simultaneous explanation support of the forbidden size. Hence each fixed support contributes at most one bad finite line parameter, and summing over supports gives the displayed bound. □

For a finite-parameter power curve

$$W_\gamma = f_0 + \gamma f_1 + \cdots + \gamma^d f_d, \quad \gamma \in \mathbb{F},$$

say that a parameter is support-wise curve-bad on S if W_γ is explained by a codeword on S , but the coefficient tuple $(f_0, \dots, f_d)$ is not simultaneously explained by $C^{\equiv(d+1)}$ on S . The no-loss curve-CA version replaces this support-wise noncontainment by the corresponding global far condition at agreement a .

Lemma 3.17 (one support pays for at most d curve parameters). *Let $C = \text{RS}[\mathbb{F}, D, k]$, let $a \geq k$, and fix a support $S \subseteq D$ with $|S| \geq a$. For a degree- d power curve W_γ , at most d finite parameters $\gamma \in \mathbb{F}$ can be support-wise curve-bad on S . The same bound holds for no-loss curve-CA parameters explained on S . Consequently any support family $\mathcal{S}$ witnesses at most $d|S|$ bad finite curve parameters.*

Proof. If $d + 1$ distinct parameters $\gamma_0, \dots, \gamma_d$ were explained on the same support S , with codewords $P_j \in C$, then Lagrange interpolation in the parameter variable gives codewords $Q_0, \dots, Q_d \in C$ such that

$$\sum_{i=0}^d \gamma_j^i Q_i = P_j \quad (0 \leq j \leq d).$$

On every point of S , the same interpolation applied to the values of W_{γ_j} gives $f_i = Q_i$ for every coefficient i . Hence $(f_0, \dots, f_d)$ is simultaneously explained by $C^{\equiv(d+1)}$ on S . This contradicts support-wise curve-noncontainment on S , and also contradicts no-loss curve-CA badness at agreement a . Thus every support contributes at most d parameters. $\square$

Lemma 3.18 (one support pays for at most one interleaved tuple). *Let $C = \text{RS}[\mathbb{F}, D, k]$, let $a \geq k$, and let $\mu \geq 1$. Fix a μ -row received word $U = (U_1, \dots, U_\mu)$. For a fixed support $S \subseteq D$ with $|S| \geq a$, there is at most one tuple $(P_1, \dots, P_\mu) \in C^\mu$ agreeing with U on every row over S . Hence the number of interleaved list tuples whose common agreement support belongs to a support family $\mathcal{S}$ is at most $|\mathcal{S}|$.*

Proof. For each row i , two degree- $< k$ polynomials agreeing with U_i on the fixed support S agree with each other on at least k points, so they are equal. Thus each row has at most one codeword compatible with S , and therefore the whole interleaved tuple is unique if it exists. $\square$

We now instantiate the support-family ledger on quotient-remainder supports. For $c \mid n$ and an integer A with $0 \leq A \leq n$, write

$$A = m_c(A)c + s_c(A), \quad 0 \leq s_c(A) < c,$$

and define

$$\begin{aligned} \mathcal{S}_c(A) := \{ & \pi_c^{-1}(M) \cup R : M \subseteq Q_c, |M| = m_c(A), \\ & R \subseteq D \setminus \pi_c^{-1}(M), |R| = s_c(A) \}. \end{aligned}$$

For a divisor set $\mathcal{C} \subseteq \{c : c \mid n\}$ put

$$\mathcal{S}_c^{\geq a} := \bigcup_{c \in \mathcal{C}} \bigcup_{A=a}^n \mathcal{S}_c(A)$$

and

$$U_{\mathcal{C}}^{\text{supp}}(a) := |\mathcal{S}_{\mathcal{C}}^{\geq a}|.$$

When the exact union is inconvenient, use the safe sum

$$U_{\mathcal{C}}^{\text{sum}}(a) := \sum_{c \in \mathcal{C}} \sum_{A=a}^n \binom{n/c}{\lfloor A/c \rfloor} \binom{n - c\lfloor A/c \rfloor}{A - c\lfloor A/c \rfloor},$$

so that $U_{\mathcal{C}}^{\text{supp}}(a) \leq U_{\mathcal{C}}^{\text{sum}}(a)$.

Proposition 3.19 (divisor-union quotient support ledger). *Let $C = \text{RS}[\mathbb{F}, D, k]$ and let $a \geq k$. Fix a divisor set $\mathcal{C}$ and let $\mathcal{S}_{\mathcal{C}}^{\geq a}$ be the quotient-remainder support family above. Then, for every received line, the number of finite support-wise MCA-bad slopes whose witness support lies in $\mathcal{S}_{\mathcal{C}}^{\geq a}$ is at most*

$$U_{\mathcal{C}}^{\text{supp}}(a) \leq U_{\mathcal{C}}^{\text{sum}}(a).$$

The same bound holds for no-loss CA slopes witnessed by this family. For a finite-parameter degree- d power curve, the corresponding support-wise curve-MCA and no-loss curve-CA numerator is at most

$$dU_{\mathcal{C}}^{\text{supp}}(a) \leq dU_{\mathcal{C}}^{\text{sum}}(a).$$

For every interleaving arity $\mu \geq 1$, the number of interleaved list tuples whose common agreement support belongs to $\mathcal{S}_{\mathcal{C}}^{\geq a}$ is at most $U_{\mathcal{C}}^{\text{supp}}(a)$.

Proof. Apply Lemmas 3.16, 3.17, and 3.18 to the support family $\mathcal{S}_{\mathcal{C}}^{\geq a}$. The inequality $U_{\mathcal{C}}^{\text{supp}}(a) \leq U_{\mathcal{C}}^{\text{sum}}(a)$ is only the union bound over the divisor and agreement-size indices. $\square$

Remark 3.20 (what this does and does not finish). Proposition 3.19 supplies a certified safe-side quotient ledger at the support-family level, and it is already enough for a staircase scanner to avoid double-counting divisor scales: one may compute the exact union $U_{\mathcal{C}}^{\text{supp}}$ or use the printed safe sum. This is not yet the sharp quotient upper theorem needed for a full prize solution. Near capacity the support count can be exponentially larger than the number of distinct bad slopes, so the next refinement is a slope-collision quotient ledger: classify when many quotient-remainder supports map to the same line or curve parameter, and replace the support count by the corresponding distinct-parameter union count.

Definition 3.21 (block support-profile polynomial). Let D be a multiplicative coset of order n , let $\mathcal{C}$ be a nonempty finite set of divisors of n , and put

$$h := \text{lcm } \mathcal{C}, \quad N_h := n/h.$$

For each $c \in \mathcal{C}$ let $H_c \leq H_h$ denote the subgroup of the cyclic kernel of order h with $|H_c| = c$. If $T \subseteq H_h$, define

$$\phi_c(T) := \#\{\text{cosets } uH_c \subseteq H_h : uH_c \subseteq T\},$$

the number of complete c -subfibers contained in the block subset T . The block support-profile polynomial of $\mathcal{C}$ is

$$G_{\mathcal{C},h}(Y, (Z_c)_{c \in \mathcal{C}}) := \sum_{T \subseteq H_h} Y^{|T|} \prod_{c \in \mathcal{C}} Z_c^{\phi_c(T)}.$$

For a vector $\nu = (\nu_c)_{c \in \mathcal{C}}$ write $Z^{\nu} := \prod_{c \in \mathcal{C}} Z_c^{\nu_c}$, and define $N_{\mathcal{C},h}(A; \nu)$ by

$$G_{\mathcal{C},h}(Y, Z)^{N_h} = \sum_{A=0}^n \sum_{\nu} N_{\mathcal{C},h}(A; \nu) Y^A Z^{\nu}.$$

Lemma 3.22 (block polynomial counts exact quotient profiles). *For a support $S \subseteq D$ and $c \mid n$, let*

$$F_c(S) := \#\{c\text{-fibers of } D \rightarrow D^c \text{ contained in } S\}.$$

Then $N_{\mathcal{C},h}(A; \nu)$ is exactly the number of supports $S \subseteq D$ such that

$$|S| = A, \quad F_c(S) = \nu_c \quad (c \in \mathcal{C}).$$

Proof. The h -fibers partition D into N_h blocks, and every c -fiber with $c \in \mathcal{C}$ is contained in a unique h -fiber because $c \mid h$. After choosing an identification of each h -fiber with the cyclic kernel H_h , the number of selected points and the number of complete internal c -subfibers are recorded by the monomial

$$Y^{|T|} \prod_{c \in \mathcal{C}} Z_c^{\phi_c(T)}$$

for the block subset T . The choices in distinct h -fibers are independent, and the global statistics $|S|$ and $F_c(S)$ are the sums of the block statistics. Multiplying the block polynomial over the N_h blocks and extracting the coefficient gives the asserted count. $\square$

Lemma 3.23 (membership by full-fiber profile). *Fix A and $c \mid n$, and write*

$$A = m_c(A)c + s_c(A), \quad 0 \leq s_c(A) < c.$$

For a support $S \subseteq D$ of size A , one has

$$S \in \mathcal{S}_c(A) \quad \Longleftrightarrow \quad F_c(S) = m_c(A) = \lfloor A/c \rfloor.$$

Proof. If $S \in \mathcal{S}_c(A)$, then by definition it is the union of $m_c(A)$ complete c -fibers and $s_c(A) < c$ residual points outside those fibers; there is no room for any further complete c -fiber. Hence $F_c(S) = m_c(A)$. Conversely, if $F_c(S) = m_c(A)$, take the complete c -fibers contained in S as the set M in the definition of $\mathcal{S}_c(A)$; the remaining $A - m_c(A)c = s_c(A)$ selected points form the residual set. $\square$

Theorem 3.24 (exact divisor-lattice support-union formula). *Let $\mathcal{C}$ be a nonempty finite set of divisors of n , and define $G_{\mathcal{C},h}$ and $N_{\mathcal{C},h}(A; \nu)$ as in Definition 3.21. Then the exact support-family union from Proposition 3.19 is*

$$U_{\mathcal{C}}^{\text{supp}}(a) = \sum_{A=a}^n \sum_{\nu} N_{\mathcal{C},h}(A; \nu) \mathbf{1}[\exists c \in \mathcal{C} : \nu_c = \lfloor A/c \rfloor].$$

Consequently the quotient-support branch numerators in Proposition 3.19 may be computed by this coefficient formula instead of by the safe sum $U_{\mathcal{C}}^{\text{sum}}(a)$.

Proof. For a fixed exact size A , the union $\bigcup_{c \in \mathcal{C}} \mathcal{S}_c(A)$ consists precisely of those supports S of size A for which $S \in \mathcal{S}_c(A)$ for at least one divisor $c \in \mathcal{C}$. By Lemma 3.23, this is exactly the condition

$$F_c(S) = \lfloor A/c \rfloor$$

for at least one $c \in \mathcal{C}$. By Lemma 3.22, the number of supports with a fixed profile vector $\nu = (F_c(S))_{c \in \mathcal{C}}$ is $N_{\mathcal{C},h}(A; \nu)$. Summing over all exact sizes $A \geq a$ and over the profile vectors satisfying the displayed predicate gives the formula. $\square$

Corollary 3.25 (single-divisor sanity check). *For $\mathcal{C} = \{c\}$ the block polynomial is*

$$G_{\{c\},c}(Y, Z) = (1 + Y)^c + (Z - 1)Y^c,$$

and Theorem 3.24 gives

$$|\mathcal{S}_c(A)| = \binom{n/c}{\lfloor A/c \rfloor} \binom{n - c\lfloor A/c \rfloor}{A - c\lfloor A/c \rfloor},$$

the summand used in the safe ledger.

Proof. Inside one c -fiber, every proper subset contributes only $Y^{|T|}$, while the full subset contributes $Y^c Z$. Thus the block polynomial is $(1 + Y)^c - Y^c + Y^c Z$. Raising to n/c and selecting exactly $m = \lfloor A/c \rfloor$ full blocks and $s = A - mc$ residual points in the remaining blocks gives the displayed count. $\square$

Definition 3.26 (divisor-block support-ledger certificate). A divisor-block support-ledger certificate for a divisor set $\mathcal{C}$ and a threshold a consists of:

- (i) the lcm block size $h = \text{lcm } \mathcal{C}$ and the block count $N_h = n/h$;
- (ii) the internal coset table of every subgroup $H_c \leq H_h$ for $c \in \mathcal{C}$;
- (iii) the block polynomial $G_{\mathcal{C},h}$, either as a full coefficient table or as an orbit-compressed table under translations of H_h ;
- (iv) the coefficient extraction from $G_{\mathcal{C},h}^{N_h}$ for every exact size $A \geq a$;
- (v) the predicate table $\mathbf{1}[\exists c \in \mathcal{C} : \nu_c = \lfloor A/c \rfloor]$ and the final integer $U_{\mathcal{C}}^{\text{supp}}(a)$.

The verifier checks the block enumeration, the subgroup containment relation $c \mid h$, coefficient arithmetic, and the final predicate sum. This is an exact support-union certificate; it does not by itself coalesce distinct line or curve parameters that arise from different supports.

Remark 3.27 (support milestone versus slope milestone). Theorem 3.24 closes the divisor-lattice overlap problem at the support-family level: intersections between quotient descriptions at incomparable divisor scales are now counted by one coefficient extraction in the common h -block. This is still weaker than the desired distinct-slope quotient ledger. To finish that stronger milestone one must apply the exact support-image maps of Propositions 3.30 and 3.32 to the support set counted here and then prove or certify the resulting parameter image size after duplicate coalescing.

3.3 Distinct-parameter quotient image ledger

The support-union ledger of Section 3.2 is deliberately safe but often very loose: many quotient-remainder supports can certify the same line slope or the same curve parameter. The next refinement is to count the image of the actual parameter map. This subsection gives an exact image ledger for the quotient branch. It is still a branch ledger, not an aperiodic upper bound: it counts only witnesses whose agreement support lies in a declared quotient-remainder support family.

We first record the support-to-parameter map in syndrome form. Let $C = \text{RS}[\mathbb{F}, D, k]$, $|D| = n$, and put $r = n - k$. For each $x \in D$ set

$$\lambda_x := \left(\prod_{y \in D, y \neq x} (x - y) \right)^{-1}.$$

For a word $Y : D \rightarrow \mathbb{F}$, define its syndrome vector

$$\text{Syn}(Y)_m := \sum_{x \in D} \lambda_x x^m Y(x), \quad 0 \leq m < r.$$

Thus $\text{Syn}(Y) = 0$ if and only if $Y \in C$. If $T \subseteq D$ has size $j \leq r$, write

$$L_T(X) = \prod_{x \in T} (X - x) = \ell_0 + \ell_1 X + \cdots + \ell_j X^j, \quad \ell_T = (\ell_0, \dots, \ell_j)^t,$$

and put $t = r - j$. For $w \in \mathbb{F}^r$ let $H_{t,j}(w)$ be the $t \times (j + 1)$ Hankel window

$$H_{t,j}(w)_{a,b} = w_{a+b}, \quad 0 \leq a < t, \quad 0 \leq b \leq j.$$

When $S = D \setminus T$, the integer t is exactly $|S| - k$.

Lemma 3.28 (support-locator syndrome recurrence). *Let $Y : D \rightarrow \mathbb{F}$ and let $T \subseteq D$ with $|T| = j \leq r$. Put $S = D \setminus T$ and $t = r - j$. Then Y is explained by C on S if and only if*

$$H_{t,j}(\text{Syn}(Y))\ell_T = 0.$$

Proof. Let $W_T \subseteq \mathbb{F}^r$ be the span of the parity-check columns indexed by T ,

$$W_T = \text{span}_{\mathbb{F}}\{\lambda_x(1, x, \dots, x^{r-1}) : x \in T\}.$$

The word Y is explained by C on S if and only if $Y - c$ is supported on T for some $c \in C$, equivalently $\text{Syn}(Y) \in W_T$. If $w \in W_T$, write $w_m = \sum_{x \in T} a_x x^m$. Then, for $0 \leq h < t$,

$$(H_{t,j}(w)\ell_T)_h = \sum_{b=0}^j \ell_b w_{h+b} = \sum_{x \in T} a_x x^h L_T(x) = 0.$$

Conversely, because L_T is monic, the displayed recurrence determines all coordinates $w_j, \dots, w_{r-1}$ from the first j coordinates, so its solution space has dimension at most j . The j parity-check columns indexed by the distinct points of T have Vandermonde rank j , hence W_T has dimension j and is exactly the recurrence space. $\square$

Lemma 3.29 (exact affine and projective support-image map). *Let $f, g : D \rightarrow \mathbb{F}$, put $u = \text{Syn}(f)$ and $v = \text{Syn}(g)$, and fix $T \subseteq D$ with $|T| = j \leq r$ and $S = D \setminus T$. Define*

$$A_T := H_{t,j}(u)\ell_T, \quad B_T := H_{t,j}(v)\ell_T.$$

A finite slope $z \in \mathbb{F}$ is explained on the fixed support S and is support-wise noncontained there if and only if

$$A_T + zB_T = 0, \quad B_T \neq 0.$$

Consequently the fixed support contributes either no finite bad slope or the unique value

$$\theta_T = -A_{T,m}/B_{T,m}$$

for any coordinate m with $B_{T,m} \neq 0$, provided all nonzero coordinates give the same value.

For the projective line $\alpha f + \beta g$, the fixed support contributes a support-wise noncontained point $[\alpha : \beta] \in \mathbb{P}^1(\mathbb{F})$ if and only if A_T and B_T are linearly dependent and not both zero. In that case the contributed projective parameter is the unique point satisfying

$$\alpha A_T + \beta B_T = 0.$$

Proof. A word Y is explained by C on $S = D \setminus T$ if and only if $Y - c$ is supported on T for some $c \in C$, which is equivalent to $\text{Syn}(Y)$ lying in the span of the parity-check columns indexed by T . The standard locator recurrence says that this is equivalent to

$$H_{t,j}(\text{Syn}(Y))\ell_T = 0.$$

Applying this to $Y = f + zg$ gives $A_T + zB_T = 0$. Under this incidence, simultaneous explanation of f and g on S is equivalent to $B_T = 0$, because $A_T = (A_T + zB_T) - zB_T$. This proves the finite-slope statement. The projective statement is the same calculation with $Y = \alpha f + \beta g$; if $A_T = B_T = 0$ then both coefficient words are already explained on S , so the point is contained rather than bad. $\square$

Let $\mathcal{S}_C^{\geq a}$ be the quotient-remainder support family from Section 3.2, and let

$$\mathcal{T}_C^{\geq a} := \{D \setminus S : S \in \mathcal{S}_C^{\geq a}\}$$

be the corresponding co-support family. For a received line define

$$\Theta_C^{\text{aff}}(f, g; a) := \{\theta_T : T \in \mathcal{T}_C^{\geq a}, \theta_T \text{ is defined by Lemma 3.29}\}$$

and define $\Theta_C^{\text{proj}}(f, g; a)$ analogously using the projective point of Lemma 3.29.

Proposition 3.30 (distinct-parameter quotient ledger for lines). *For every received line $f + zg$, the finite affine slopes that are support-wise MCA-bad and have a witness support in $\mathcal{S}_C^{\geq a}$ are exactly the elements of $\Theta_C^{\text{aff}}(f, g; a)$. Hence the exact quotient-branch line numerator is*

$$|\Theta_C^{\text{aff}}(f, g; a)| \leq U_C^{\text{supp}}(a).$$

The corresponding projective-slope numerator is

$$|\Theta_C^{\text{proj}}(f, g; a)| \leq U_C^{\text{supp}}(a).$$

The same image ledgers upper-bound no-loss CA parameters whose explaining support lies in the same quotient family.

Proof. This is only Lemma 3.29 applied to every co-support in the family, followed by taking a union of the resulting parameter values. The CA claim follows because no-loss CA-badness is stronger than having an explained slope on the support and forbids simultaneous explanation on any support of size at least a . $\square$

The curve analogue is just as explicit. For a degree- d power curve

$$W_\gamma = f_0 + \gamma f_1 + \cdots + \gamma^d f_d, \quad \gamma \in \mathbb{F},$$

put $u_i = \text{Syn}(f_i)$ and, for $T \subseteq D$, define

$$V_{i,T} := H_{t,j}(u_i) \ell_T \in \mathbb{F}^t.$$

Let

$$\Gamma_T^{(d)} := \{\gamma \in \mathbb{F} : V_{0,T} + \gamma V_{1,T} + \cdots + \gamma^d V_{d,T} = 0, (V_{0,T}, \dots, V_{d,T}) \neq 0\}.$$

The nonzero tuple condition is exactly support-wise noncontainment of the coefficient tuple on the fixed support.

Lemma 3.31 (exact support-image map for finite-parameter power curves). *For a fixed support $S = D \setminus T$, the set of support-wise noncontained curve-bad finite parameters is exactly $\Gamma_T^{(d)}$. In particular $|\Gamma_T^{(d)}| \leq d$.*

Proof. The same syndrome-locator recurrence applied to W_γ gives the displayed vector equation. If every $V_{i,T}$ is zero, then every coefficient word f_i is already explained on S , so the parameter is contained. Otherwise at least one scalar coordinate of $V_{0,T} + \gamma V_{1,T} + \cdots + \gamma^d V_{d,T}$ is a nonzero polynomial in γ of degree at most d , so it has at most d roots. $\square$

Define the exact quotient curve image

$$\Gamma_C^{(d)}(f_0, \dots, f_d; a) := \bigcup_{T \in \mathcal{T}_C^{\geq a}} \Gamma_T^{(d)}.$$

Proposition 3.32 (distinct-parameter quotient ledger for curves). *For a finite-parameter degree- d power curve, the quotient-branch support-wise curve-MCA numerator is exactly*

$$|\Gamma_C^{(d)}(f_0, \dots, f_d; a)|.$$

It satisfies the safe bounds

$$|\Gamma_C^{(d)}(f_0, \dots, f_d; a)| \leq d U_C^{\text{supp}}(a) \leq d U_C^{\text{sum}}(a).$$

The same image ledger upper-bounds no-loss curve-CA parameters whose explaining support lies in the quotient family.

Proof. Apply Lemma 3.31 to each support and take the union of the resulting parameter sets. The displayed inequalities follow from $|\Gamma_T^{(d)}| \leq d$ and from the support-union bound of Proposition 3.19. $\square$

Definition 3.33 (fixed-support parameter gcd). Let $C = \text{RS}[\mathbb{F}, D, k]$, let $q = |\mathbb{F}|$, and let

$$W_\Gamma = f_0 + \Gamma f_1 + \cdots + \Gamma^d f_d, \quad \Gamma \in \mathbb{F},$$

be a finite-parameter degree- d power curve of received words. Fix a co-support $T \subseteq D$, put $S = D \setminus T$, and assume $|S| > k$. With the notation of Lemma 3.31, write

$$P_{T,m}(\Gamma) := \sum_{i=0}^d \Gamma^i (V_{i,T})_m, \quad 0 \leq m < t, \quad t = |S| - k.$$

Define the fixed-support contribution polynomial $g_T^{(d)}(\Gamma) \in \mathbb{F}[\Gamma]$ as follows. If all coordinate polynomials $P_{T,m}$ are identically zero, set $g_T^{(d)} = 1$. Otherwise let $g_T^{(d)}$ be the monic greatest common divisor of the nonzero polynomials among $P_{T,0}, \dots, P_{T,t-1}$.

Lemma 3.34 (fixed-support gcd equals the noncontained parameter set). *In the setting of Definition 3.33, the finite parameters $\gamma \in \mathbb{F}$ for which W_γ is explained on $S = D \setminus T$ and the coefficient tuple $(f_0, \dots, f_d)$ is not simultaneously explained on S are exactly the roots in $\mathbb{F}$ of $g_T^{(d)}$. In particular*

$$\deg g_T^{(d)} \leq d,$$

with equality possible only before taking common factors across different supports.

Proof. By Lemma 3.31, the fixed support explains W_γ exactly when

$$P_{T,m}(\gamma) = 0 \quad (0 \leq m < t).$$

If every coordinate polynomial is identically zero, then every coefficient vector $V_{i,T}$ is zero, so every coefficient word f_i is already explained on S ; this is the contained branch and contributes no noncontained parameter. The convention $g_T^{(d)} = 1$ gives the empty root set, as required.

Otherwise the common zeros of the coordinate polynomials are exactly the roots of their monic gcd after zero coordinates are discarded. Since at least one coordinate polynomial is nonzero of degree at most d , the gcd also has degree at most d . The nonzero tuple condition in Lemma 3.31 is precisely the assertion that the coordinate polynomials are not all zero. $\square$

Definition 3.35 (quotient image lcm polynomial). Let $\mathcal{T}$ be any finite family of co-supports $T \subseteq D$ with $|D \setminus T| > k$. For a degree- d finite-parameter power curve define

$$R_{\mathcal{T}}^{(d)}(\Gamma) := \text{rad lcm}_{T \in \mathcal{T}} g_T^{(d)}(\Gamma),$$

where factors equal to 1 are ignored and the empty lcm is 1. For the quotient branch at agreement threshold $a > k$ and divisor set $\mathcal{C}$, we use

$$\mathcal{T} = \mathcal{T}_{\mathcal{C}}^{\geq a} := \{D \setminus S : S \in \mathcal{S}_{\mathcal{C}}^{\geq a}\}.$$

The finite-field root-count polynomial is

$$R_{\mathcal{T},F}^{(d)}(\Gamma) := \gcd(R_{\mathcal{T}}^{(d)}(\Gamma), \Gamma^q - \Gamma).$$

Because $R_{\mathcal{T}}^{(d)}$ is squarefree, the degree of $R_{\mathcal{T},F}^{(d)}$ is the number of $\mathbb{F}$ -rational roots of $R_{\mathcal{T}}^{(d)}$.

Theorem 3.36 (exact finite-parameter quotient image ledger). *Let $a > k$, let $\mathcal{C}$ be a finite divisor set, and let $\mathcal{T}_{\mathcal{C}}^{\geq a}$ be the quotient co-support family. For every finite-parameter degree- d power curve, the exact quotient-branch support-wise curve-MCA numerator is*

$$\boxed{\left| \Gamma_{\mathcal{C}}^{(d)}(f_0, \dots, f_d; a) \right| = \deg R_{\mathcal{T}_{\mathcal{C}}^{\geq a}, F}^{(d)}}.$$

Consequently

$$\left| \Gamma_{\mathcal{C}}^{(d)}(f_0, \dots, f_d; a) \right| \leq \deg R_{\mathcal{T}_{\mathcal{C}}^{\geq a}}^{(d)} \leq d U_{\mathcal{C}}^{\text{supp}}(a),$$

and the final support count may be evaluated by the exact divisor-block formula of Theorem 3.24. The same lcm ledger upper-bounds no-loss curve-CA parameters whose explaining support lies in the same quotient family.

Proof. By Lemma 3.34, for each fixed co-support T the noncontained parameters witnessed on $D \setminus T$ are exactly the $\mathbb{F}$ -roots of $g_T^{(d)}$. Taking the union over $T \in \mathcal{T}_C^{\geq a}$ is therefore the same as taking the $\mathbb{F}$ -rational root set of the squarefree lcm $R_{\mathcal{T}_C^{\geq a}}^{(d)}$. Over a finite field, the $\mathbb{F}$ -rational roots of a squarefree polynomial are counted by the degree of its gcd with $\Gamma^q - \Gamma$, proving the displayed equality.

The degree bound follows from $\deg g_T^{(d)} \leq d$ and from the fact that a squarefree lcm has degree at most the sum of the degrees of the factors before coalescing. The no-loss curve-CA claim follows because a no-loss CA parameter with an explaining support in the quotient family is, on that support, noncontained; otherwise the coefficient tuple would have a simultaneous explanation on a support of size at least a . $\square$

Corollary 3.37 (affine lines as the degree-one lcm ledger). *For an affine received line $f + Zg$, take $d = 1$, $f_0 = f$, and $f_1 = g$. Then $R_{\mathcal{T}_C^{\geq a}}^{(1)}(Z)$ is the squarefree product of the linear factors $Z - z$ for the distinct finite quotient-witnessed support-wise MCA-bad slopes. Hence*

$$\#\{\text{finite quotient-witnessed MCA-bad slopes}\} = \deg R_{\mathcal{T}_C^{\geq a}}^{(1)}.$$

The same polynomial upper-bounds no-loss CA finite slopes whose explaining support lies in the declared quotient family.

Proof. For $d = 1$, each nonconstant fixed-support gcd has degree at most one. Its root, if present, is exactly the unique finite slope supplied by Lemma 3.29. Taking the squarefree lcm over the declared co-support family therefore multiplies one copy of $Z - z$ for each distinct finite slope and no copies for supports that contribute no finite noncontained slope. $\square$

Definition 3.38 (projective line lcm ledger). For a projective received line $Z_0f + Z_1g$ and a co-support T with $|D \setminus T| > k$, form the homogeneous linear coordinate forms

$$L_{T,m}(Z_0, Z_1) := Z_0(A_T)_m + Z_1(B_T)_m, \quad 0 \leq m < t.$$

If all $L_{T,m}$ are zero, set $h_T^{\text{proj}} = 1$. Otherwise let h_T^{proj} be the gcd of the nonzero linear forms, normalized by a fixed monomial order. Thus h_T^{proj} is either 1 or a homogeneous linear form. For a co-support family $\mathcal{T}$ define

$$R_{\mathcal{T}}^{\text{proj}}(Z_0, Z_1) := \text{rad lcm}_{T \in \mathcal{T}} h_T^{\text{proj}}.$$

Corollary 3.39 (exact projective quotient image ledger). *For projective slopes, the exact quotient-branch support-wise MCA numerator is*

$$\deg R_{\mathcal{T}_C^{\geq a}}^{\text{proj}}.$$

The same degree upper-bounds no-loss projective-CA parameters whose explaining support lies in the quotient family.

Proof. A nonzero family of homogeneous linear forms on $\mathbb{P}^1$ has a common zero if and only if the forms have a common homogeneous linear factor; this factor cuts out exactly one $\mathbb{F}$ -rational projective point. The all-zero case is the contained branch and contributes no bad point, by convention. The squarefree lcm therefore contains one linear factor for each distinct projective parameter and no duplicates. $\square$

Definition 3.40 (lcm quotient-image certificate). An lcm quotient-image certificate for a line or finite-parameter curve consists of:

- (i) the divisor set $\mathcal{C}$, the agreement threshold a , and either the exact support-union certificate of Definition 3.26 or an explicit hash/orbit enumeration of the co-support family $\mathcal{T}_{\mathcal{C}}^{\geq a}$;
- (ii) for every represented co-support T , the locator vector ℓ_T and the syndrome-window coefficient vectors $V_{i,T}$;
- (iii) the coordinate polynomials $P_{T,m}(\Gamma)$, the fixed-support gcd $g_T^{(d)}$, and a proof that the all-zero case has been assigned to the contained branch;
- (iv) the squarefree lcm $R_{\mathcal{T}}^{(d)}$ and, over the finite field, the root-count factor $R_{\mathcal{T},F}^{(d)} = \gcd(R_{\mathcal{T}}^{(d)}, \Gamma^q - \Gamma)$;
- (v) the declared numerator $\deg R_{\mathcal{T},F}^{(d)}$ for finite curves or $\deg R_{\mathcal{T}}^{\text{proj}}$ for projective lines.

The verifier checks the locator split condition, the syndrome recurrence, the gcd and lcm arithmetic, squarefreeness, and the final finite-field root count. This certificate replaces a support count by an exact distinct-parameter count inside the declared quotient branch.

Remark 3.41 (status after the lcm image milestone). The quotient ledgers now separate three tasks that were previously entangled. First, Theorem 3.8 supplies lower floors by heaviest generated-field prefix fibers. Second, Theorem 3.24 counts the declared quotient support union with divisor overlaps handled exactly. Third, Theorem 3.36 and Corollaries 3.37 and 3.39 push that support family through the fixed-locator equations and coalesce duplicate finite, projective, and curve parameters by gcd/lcm arithmetic. The remaining quotient-side obstruction is not duplicate coalescing inside a declared quotient family; it is the structural safe-side theorem that every bad parameter in the intended regime either belongs to this quotient image, to the tangent branch, or to the aperiodic/extension buckets bounded elsewhere.

Definition 3.42 (quotient image certificate). A quotient image certificate for a line or curve ledger consists of:

- (i) the divisor set $\mathcal{C}$ and the agreement threshold a ;
- (ii) a deterministic description, hash, or orbit decomposition of $\mathcal{T}_{\mathcal{C}}^{\geq a}$;
- (iii) for every represented co-support T , the locator coefficients ℓ_T and the syndrome-window vectors A_T, B_T for a line, or $V_{0,T}, \dots, V_{d,T}$ for a curve;
- (iv) the declared parameter value, parameter bucket, or the declaration that no parameter is contributed;
- (v) a bucket table proving the number of distinct parameters after duplicate coalescing.

The verifier checks the locator split condition, the Hankel equations, the noncontainment vector, and the bucket consistency. This certificate is the preferred replacement for a raw quotient-support count whenever the support count is much larger than the actual parameter image.

Remark 3.43 (status of the refinement). The propositions above are exact for the declared quotient branch and are a real refinement over the support ledger: the numerator is an image size, not a support count. They still do not prove that all bad parameters are quotient-periodic. The remaining safe-side theorem for the full prize is the aperiodic Hankel-packing bound: after tangent and quotient image branches are removed, one must bound the parameters coming from co-supports outside the declared quotient families.

3.4 Aperiodic Hankel chart atlas

The quotient image ledger of Section 3.3 counts bad parameters whose witness support lies in a declared quotient-remainder family. The complementary safe-side problem is to control all other supports. This subsection is the contributor-facing version of that problem. It does not claim that the needed aperiodic eliminants always exist. It gives a finite chart atlas, with precise ideals and pivots, so that a verifier can check a supplied eliminant and convert it into a bad-parameter upper bound.

The main point is to avoid one huge saturated incidence ideal. We split the aperiodic branch into:

- (i) a regular overdetermined Hankel bucket, where a single maximal minor in the line parameter already gives an eliminant;
- (ii) finite affine pivot charts, one for each noncontainment coordinate;
- (iii) a separate projective-infinity chart, if projective slopes are being counted;
- (iv) finite-parameter curve pivot charts, one for a nonzero coefficient of one coordinate polynomial;
- (v) a singular bucket, where all regular minors vanish or the projection to the parameter line remains positive-dimensional after the previous splits.

Only the last bucket is genuinely unresolved. A staircase scanner should keep it as a named obstruction rather than hiding it inside the word “aperiodic.”

Fix an exact agreement value $A \geq k$ and put

$$j = j_A := n - A, \quad t = t_A := A - k, \quad R = n - k.$$

Let

$$P_D(X) := \prod_{x \in D} (X - x).$$

A monic degree- j co-support locator is written

$$L_\ell(X) = X^j + \ell_{j-1}X^{j-1} + \cdots + \ell_0, \quad \ell = (\ell_0, \dots, \ell_{j-1}) \in \mathbb{A}^j.$$

Let $R_{D,j}(\ell; X)$ be the remainder of $P_D(X)$ modulo $L_\ell(X)$, and let $I_{\text{split},j}$ be the ideal generated by the j coefficients of $R_{D,j}$. Since P_D is squarefree, the F -points of $V(I_{\text{split},j})$ are exactly the squarefree D -split monic degree- j locators.

For a word $y : D \rightarrow F$ write $\text{Syn}(y) \in F^R$ for the parity-check syndrome used above. Let

$$\tilde{\ell} = (\ell_0, \dots, \ell_{j-1}, 1)^t.$$

For an affine line $f + zg$, put $u = \text{Syn}(f)$ and $v = \text{Syn}(g)$ and define

$$A(\ell) := H_{t,j}(u)\tilde{\ell}, \quad B(\ell) := H_{t,j}(v)\tilde{\ell} \quad \in F^t.$$

For a fixed split locator, the support-wise finite-slope condition is exactly

$$A(\ell) + ZB(\ell) = 0, \quad B(\ell) \neq 0.$$

The inequality is the noncontainment condition.

3.4.1 Regular overdetermined bucket

Assume first that

$$t_A \geq j_A + 1, \quad \text{equivalently} \quad 2A \geq n + k + 1.$$

Set

$$M_A(Z) := H_{t_A, j_A}(u) + ZH_{t_A, j_A}(v).$$

This is a $t_A \times (j_A + 1)$ matrix with entries affine-linear in Z .

Definition 3.44 (regular Hankel minor certificate). A regular Hankel minor certificate at exact agreement A is a row set $I_A \subseteq \{0, \dots, t_A - 1\}$ of size $j_A + 1$ such that

$$\Delta_A(Z) := \det M_A(Z)_{I_A, \{0, \dots, j_A\}}$$

is not the zero polynomial in $F[Z]$. If no such row set exists, the line is called *Hankel-singular* at A .

Lemma 3.45 (regular exact-agreement eliminant). *Assume $t_A \geq j_A + 1$, and suppose $\Delta_A(Z)$ is a regular Hankel minor certificate. Every finite support-wise MCA-bad slope with a witness support of exact size A is a root of Δ_A . Hence the number of such slopes is at most*

$$\deg \Delta_A \leq j_A + 1 = n - A + 1.$$

The same root containment holds for no-loss CA-bad slopes with exact agreement A .

Proof. Let z be a bad finite slope with witness support S of size A , and put $T = D \setminus S$. Then $|T| = j_A$, and the locator vector $\tilde{\ell}_T$ is a nonzero kernel vector of $M_A(z)$. Therefore the column rank of $M_A(z)$ is less than $j_A + 1$, so every $(j_A + 1) \times (j_A + 1)$ maximal minor vanishes at z , including $\Delta_A(z)$. The degree bound follows because Δ_A is the determinant of a $(j_A + 1) \times (j_A + 1)$ matrix whose entries are affine-linear in Z . $\square$

Theorem 3.46 (regular closed-range Hankel packing). *Let $a \geq k$ and suppose that for each exact agreement $A \in \{a, a + 1, \dots, n\}$ with $t_A \geq j_A + 1$ a regular Hankel minor certificate Δ_A is supplied. Then all bad finite slopes whose exact witness size is in this regular bucket lie in the root set of*

$$\Delta_{\geq a}(Z) := \prod_{A=a}^n \Delta_A(Z),$$

with factors omitted for exact agreements outside the overdetermined range. Consequently their number is at most

$$\sum_A \deg \Delta_A.$$

If exact root sets are supplied, roots already charged to the tangent or quotient-image ledgers may be removed from this count by duplicate coalescing.

Proof. Apply Lemma 3.45 at the exact size of the witness support and take the union of the resulting root sets. The degree sum is a safe union bound. $\square$

Canonical regular rank-drop gcds. The regular-minor certificate below can be made canonical. Instead of choosing one nonzero maximal minor, take the greatest common divisor of all nonzero maximal minors. This gives exactly the finite-parameter values at which the Hankel pencil can drop rank, before the split-locator and noncontainment tests are imposed. Thus it is an upper ledger for the regular bucket, and often a much smaller one than the degree of a single selected minor.

Fix an exact agreement value $A \geq k$, and keep the notation

$$j = j_A := n - A, \quad t = t_A := A - k, \quad r = n - k.$$

The regular overdetermined range is

$$t_A > j_A, \quad \text{i.e.} \quad t_A \geq j_A + 1, \quad \text{equivalently} \quad 2A \geq n + k + 1.$$

Definition 3.47 (canonical affine rank-drop gcd). Fix an affine received line $f + Zg$, put $u = \text{Syn}(f)$ and $v = \text{Syn}(g)$, and assume $t_A \geq j_A + 1$. Set

$$M_A(Z) := H_{t_A, j_A}(u) + ZH_{t_A, j_A}(v).$$

Let $\mathfrak{M}_A(Z)$ be the finite set of all $(j_A + 1) \times (j_A + 1)$ maximal minors of $M_A(Z)$. If every element of $\mathfrak{M}_A(Z)$ is the zero polynomial, the exact-agreement bucket is called *rank-drop singular*. Otherwise define

$$G_A^{\text{aff}}(Z) := \text{rad gcd}\{\Delta(Z) : \Delta \in \mathfrak{M}_A(Z), \Delta \neq 0\},$$

with the gcd normalized to be monic. Over the finite field $\mathbb{F}$ of size q set

$$G_{A,F}^{\text{aff}}(Z) := \text{gcd}(G_A^{\text{aff}}(Z), Z^q - Z).$$

Thus $G_{A,F}^{\text{aff}}$ is the squarefree polynomial whose roots are precisely those F -rational parameters at which all nonzero maximal minors vanish.

Theorem 3.48 (canonical exact-agreement rank-drop ledger). Assume $t_A \geq j_A + 1$ and the exact-agreement bucket is not rank-drop singular. Every finite support-wise MCA-bad slope with an exact witness support of size A is a root of G_A^{aff} . Consequently the number of such slopes is at most

$$\deg G_{A,F}^{\text{aff}} \leq \deg G_A^{\text{aff}} \leq j_A + 1 = n - A + 1.$$

The same bound holds for no-loss CA-bad slopes with exact agreement A .

Proof. Let $z \in F$ be support-wise bad with exact witness support S of size A , and put $T = D \setminus S$. By Lemma 3.28, the locator vector ℓ_T is a nonzero kernel vector of $M_A(z)$. Therefore $\text{rank } M_A(z) < j_A + 1$, so every maximal minor of $M_A(Z)$ vanishes at $Z = z$. In particular every nonzero maximal minor vanishes at z , and hence their gcd G_A^{aff} vanishes at z .

The finite-field root count is exactly $\deg G_{A,F}^{\text{aff}}$ because $G_{A,F}^{\text{aff}}$ is squarefree and divides $Z^q - Z$. Each maximal minor has degree at most $j_A + 1$, since it is the determinant of a matrix whose entries are affine-linear in Z ; the gcd cannot have larger degree. The no-loss CA claim is identical, because no-loss CA-badness with exact agreement A also supplies an explaining support of size A and hence the same locator-kernel condition. $\square$

Remark 3.49 (why this improves the one-minor certificate). Lemma 3.45 used one certified nonzero maximal minor and therefore counted the roots of a single determinant. The polynomial G_A^{aff} uses all maximal minors and counts their common finite-field root set. It is never worse than any one-minor certificate and can be much smaller; it is also canonical, so two scanners that use the same syndrome convention print the same regular-bucket polynomial.

Definition 3.50 (closed-ball rank-drop lcm). For an agreement threshold $a > k$, let

$$\mathcal{A}_{\text{reg}}(a) := \{A \in \{a, a+1, \dots, n\} : t_A \geq j_A + 1 \text{ and the bucket } A \text{ is not rank-drop singular}\}.$$

Define

$$G_{\geq a}^{\text{aff}}(Z) := \text{rad lcm}_{A \in \mathcal{A}_{\text{reg}}(a)} G_A^{\text{aff}}(Z), \quad G_{\geq a, F}^{\text{aff}}(Z) := \text{gcd}(G_{\geq a}^{\text{aff}}(Z), Z^q - Z),$$

with the empty lcm equal to 1. Exact agreement values $A \in \{a, \dots, n\} \setminus \mathcal{A}_{\text{reg}}(a)$ are recorded as residual buckets: they are either underdetermined ($t_A < j_A + 1$) or rank-drop singular.

Corollary 3.51 (canonical regular closed-ball Hankel packing). *All finite support-wise MCA-bad slopes whose exact witness size lies in $\mathcal{A}_{\text{reg}}(a)$ are roots of $G_{\geq a}^{\text{aff}}$. Hence their number is at most*

$$\deg G_{\geq a, F}^{\text{aff}} \leq \sum_{A \in \mathcal{A}_{\text{reg}}(a)} (n - A + 1).$$

The same bound holds for no-loss CA-bad slopes in the same regular buckets.

Proof. Apply Theorem 3.48 at the exact agreement size of the witness and take the union over A . The squarefree lcm coalesces duplicate roots across exact-agreement buckets; the final gcd with $Z^q - Z$ counts only finite-field roots. The displayed degree sum is the safe bound obtained before coalescing. $\square$

Definition 3.52 (paid-root removal for the regular branch). Suppose a squarefree polynomial $P_{\text{paid}, F}(Z)$ is supplied whose roots are exactly the finite-field parameters already charged to tangent/common-line and quotient-image ledgers. Define the regular residual polynomial

$$G_{\text{reg} \setminus \text{paid}, F}^{\text{aff}}(Z) := \frac{G_{\geq a, F}^{\text{aff}}(Z)}{\text{gcd}(G_{\geq a, F}^{\text{aff}}(Z), P_{\text{paid}, F}(Z))}.$$

Its degree is the number of regular rank-drop parameters not already paid by the previous ledgers.

Corollary 3.53 (quotient/tangent-removed regular aperiodic numerator). *If $P_{\text{paid}, F}$ is exact in the sense of Definition 3.52, then the number of finite slopes in the regular overdetermined branch that remain after removing the paid quotient/tangent parameters is at most*

$$\deg G_{\text{reg} \setminus \text{paid}, F}^{\text{aff}}.$$

This is a safe aperiodic numerator for the regular buckets. The underdetermined and rank-drop singular exact-agreement buckets must still be handled by pivot charts or by a separately named residual obstruction.

Proof. By Corollary 3.51, every regular-bucket bad slope is a root of $G_{\geq a, F}^{\text{aff}}$. Removing the roots of the exact paid polynomial deletes precisely those already charged parameters from this finite root set. The remaining root set is counted by the displayed quotient because all polynomials are squarefree divisors of $Z^q - Z$. $\square$

Definition 3.54 (finite-parameter curve rank-drop gcd). For a finite-parameter degree- d curve

$$W_\Gamma = f_0 + \Gamma f_1 + \cdots + \Gamma^d f_d,$$

put $u_i = \text{Syn}(f_i)$ and, at exact agreement A , define

$$M_A^{(d)}(\Gamma) := \sum_{i=0}^d \Gamma^i H_{t_A, j_A}(u_i).$$

When $t_A \geq j_A + 1$ and not all maximal minors of $M_A^{(d)}$ vanish identically, set

$$G_A^{(d)}(\Gamma) := \text{rad gcd}\{\Delta(\Gamma) : \Delta \text{ is a nonzero maximal minor of } M_A^{(d)}\},$$

and

$$G_{A, F}^{(d)}(\Gamma) := \text{gcd}(G_A^{(d)}(\Gamma), \Gamma^q - \Gamma).$$

Theorem 3.55 (canonical curve rank-drop ledger). *In a non-singular overdetermined exact-agreement bucket, every finite curve-MCA-bad parameter is a root of $G_A^{(d)}$. Consequently the number of such parameters is at most*

$$\deg G_{A, F}^{(d)} \leq \deg G_A^{(d)} \leq d(j_A + 1).$$

The same bound holds for no-loss curve-CA parameters in the same bucket. The closed-ball version is obtained by taking the squarefree lcm of the polynomials $G_A^{(d)}$ over the regular exact-agreement buckets and then taking the gcd with $\Gamma^q - \Gamma$.

Proof. A curve-bad parameter with exact witness support S gives, for $T = D \setminus S$, a nonzero locator vector in the kernel of $M_A^{(d)}(\gamma)$. Hence every maximal minor vanishes at γ , and the same gcd argument as in Theorem 3.48 applies. Each matrix entry has degree at most d in Γ , so each maximal minor has degree at most $d(j_A + 1)$. $\square$

Definition 3.56 (regular rank-drop certificate). A regular rank-drop certificate for a row, a received affine line or finite curve, and an agreement threshold a consists of:

- (i) the syndrome vectors of the coefficient words and the exact-agreement range $A \in \{a, \dots, n\}$;
- (ii) for every overdetermined exact agreement A , either a declaration that all maximal minors vanish identically, or the canonical gcd polynomial G_A^{aff} , respectively $G_A^{(d)}$, together with ideal-membership or raw minor arithmetic proving that it is the gcd of all nonzero maximal minors;
- (iii) the squarefree lcm over all non-singular regular buckets and the finite-root factor obtained by gcd with $Z^q - Z$ or $\Gamma^q - \Gamma$;
- (iv) if paid branches are removed, the exact paid-root polynomial and the finite root-set quotient of Definition 3.52;

- (v) a residual-bucket table listing all underdetermined and rank-drop singular exact agreements that are not covered by the regular gcd ledger.

The verifier checks the syndrome convention, the Hankel matrices, the maximal minor gcds, squarefree lcms, finite-field root factors, paid-root removal, and the residual-bucket table.

Remark 3.57 (status after the rank-drop milestone). The regular overdetermined part of the aperiodic Hankel problem is now canonical: one no longer has to choose a minor or accept a degree sum when the common root set of all maximal minors is smaller. This still does not solve the near-capacity underdetermined buckets, which are the hard part of T2 in the prize program. Those buckets remain assigned to the affine pivot charts, projective-infinity charts, curve coefficient pivots, or named singular residual obstructions.

3.4.2 Finite affine pivot charts

The regular minor test is intentionally coarse and only works directly in the overdetermined range. The next atlas is the one contributors should use to construct actual aperiodic eliminants.

Definition 3.58 (aperiodic locator chart). An aperiodic locator chart at exact agreement A is a pair

$$X = (E_X, \Delta_X),$$

where E_X is a finite list of polynomial equations in the locator coefficients ℓ , and $\Delta_X(\ell)$ is a product of declared nonzero tests. The chart represents the constructible set

$$V(I_{\text{split},j} + \langle E_X \rangle) \setminus V(\Delta_X).$$

The equations and inequations must include, or reference, the removal of already paid tangent/common-code-line and quotient-image branches. A family of charts $\mathfrak{X}_A$ is a valid aperiodic chart cover if the union of their constructible sets contains every remaining split locator of exact co-support size j_A not assigned to those earlier ledgers.

For a coordinate $h \in \{0, \dots, t-1\}$ define the affine pivot consistency ideal

$$\begin{aligned} J_{X,h}^{\text{aff}} &:= I_{\text{split},j} + \langle E_X \rangle + \langle A_m(\ell)B_h(\ell) - A_h(\ell)B_m(\ell) : 0 \leq m < t \rangle \\ &\quad + \langle ZB_h(\ell) + A_h(\ell) \rangle \subseteq F[\ell_0, \dots, \ell_{j-1}, Z], \end{aligned}$$

and saturate it by the pivot denominator:

$$\widehat{J}_{X,h}^{\text{aff}} := J_{X,h}^{\text{aff}} : (\Delta_X B_h)^\infty.$$

The equations $A_m B_h - A_h B_m = 0$ force A and B to be collinear on the pivot chart, and $ZB_h + A_h = 0$ records the unique finite slope.

Theorem 3.59 (pivoted affine-line eliminant certificate). *Let X be an aperiodic locator chart and let h be a noncontainment pivot. If there is a nonzero polynomial*

$$Q_{X,h}(Z) \in \widehat{J}_{X,h}^{\text{aff}} \cap F[Z]$$

of degree $D_{X,h}$, then the number of finite support-wise MCA-bad slopes in that chart with $B_h \neq 0$ is at most $\min\{|F|, D_{X,h}\}$. The same bound holds for no-loss CA-bad slopes in that pivot chart.

Proof. A bad slope in the chart has a split locator with $\Delta_X \neq 0$ and, on this pivot chart, $B_h \neq 0$. The Hankel incidence equation gives $A + ZB = 0$, hence the point (ℓ, Z) satisfies all generators of $J_{X,h}^{\text{aff}}$ and survives the saturation. Therefore $Q_{X,h}(Z) = 0$. A nonzero degree- $D_{X,h}$ polynomial over F has at most $D_{X,h}$ roots. $\square$

Remark 3.60 (why the pivot split is useful). The saturation by the whole vector B in the unsplit incidence ideal is replaced by the explicit open cover $B_h \neq 0$. A scanner can try all h , produce small univariate eliminants on successful pivots, and report the residual locus $B_0 = \dots = B_{t-1} = 0$ as contained for support-wise MCA. For no-loss CA, that residual locus also contributes no bad slope at the same support, because if $A = B = 0$ then both f and g are explained on the witness support.

3.4.3 Projective-slope split

For projective parameters $[Z_0 : Z_1]$ the incidence equation is

$$Z_0 A(\ell) + Z_1 B(\ell) = 0.$$

The finite patch $Z_0 \neq 0$ is exactly the affine pivot atlas above with $Z = Z_1/Z_0$. The extra point at infinity is $[0 : 1]$.

Definition 3.61 (projective infinity chart). For a chart X , the projective-infinity chart is the constructible locus

$$I_{\text{split},j} + \langle E_X \rangle + \langle B_0, \dots, B_{t-1} \rangle, \quad \Delta_X \neq 0, \quad A \neq 0.$$

If it is nonempty, it contributes the single projective parameter $[0 : 1]$. If it is empty, the infinity point contributes nothing on X .

Corollary 3.62 (projective eliminant certificate). *For a chart cover $\mathfrak{X}_A$, the projective-slope numerator is bounded by the union of the finite-patch root sets from Theorem 3.59 together with at most one extra point $[0 : 1]$, provided the infinity charts are certified either empty or nonempty. Equivalently, a homogeneous nonzero eliminant*

$$Q_X(Z_0, Z_1)$$

of degree D_X bounds the number of projective bad parameters in that chart by D_X .

Proof. The finite patch is the affine theorem. The complement of that patch is the single point $[0 : 1]$, which is governed by Definition 3.61. For the homogeneous formulation, every bad projective parameter is a zero of the homogeneous eliminant, and a nonzero homogeneous polynomial of degree D_X on $\mathbb{P}^1(F)$ has at most D_X zeros. $\square$

3.4.4 Finite-parameter curve charts

For a degree- d finite power curve

$$W_\Gamma = f_0 + \Gamma f_1 + \dots + \Gamma^d f_d,$$

put

$$V_i(\ell) := H_{t,j}(\text{Syn}(f_i)) \tilde{\ell} \in F^t, \quad 0 \leq i \leq d.$$

The curve incidence equation is

$$V_0(\ell) + \Gamma V_1(\ell) + \cdots + \Gamma^d V_d(\ell) = 0.$$

Curve-MCA noncontainment means that not all coefficient vectors $V_i(\ell)$ are zero. Thus the noncontained branch is covered by coefficient pivots (i, h) with $(V_i)_h \neq 0$.

For a chart X and a coefficient pivot (i, h) , define

$$J_{X,i,h}^{\text{curve,d}} := I_{\text{split},j} + \langle E_X \rangle + \left\langle \sum_{r=0}^d \Gamma^r (V_r(\ell))_m : 0 \leq m < t \right\rangle \subseteq F[\ell_0, \dots, \ell_{j-1}, \Gamma],$$

and set

$$\widehat{J}_{X,i,h}^{\text{curve,d}} := J_{X,i,h}^{\text{curve,d}} : (\Delta_X(V_i)_h)^\infty.$$

Theorem 3.63 (pivoted curve eliminant certificate). *If a chart X and coefficient pivot (i, h) have a nonzero eliminant*

$$Q_{X,i,h}(\Gamma) \in \widehat{J}_{X,i,h}^{\text{curve,d}} \cap F[\Gamma]$$

of degree $D_{X,i,h}$, then the number of finite degree- d curve-MCA bad parameters in that chart and pivot is at most $\min\{|F|, D_{X,i,h}\}$. The same bound applies to no-loss curve-CA parameters in the same pivot chart.

Proof. Every bad curve parameter on the pivot chart gives a split locator satisfying all curve-incidence equations and the nonzero pivot condition $(V_i)_h \neq 0$. It therefore survives the saturation and is a root of the supplied eliminant. $\square$

3.4.5 Fallback and final certificate

The eliminant certificates above are strongest when the projection to the parameter line is finite. A failed eliminant search should not be discarded; it should be split further or recorded as a singular bucket. The following fallback is safe but usually too weak for a final 2^{-128} threshold.

Proposition 3.64 (dimension-degree fallback). *Let $Y \subseteq \mathbb{A}_F^N$ be an affine variety of degree at most Δ and Krull dimension at most e . Then*

$$|Y(F)| \leq \Delta |F|^e.$$

Consequently, if a saturated Hankel incidence chart for line or curve parameters has degree Δ_X and dimension e_X , then the number of bad parameters in that chart is at most $\Delta_X |F|^{e_X}$.

Proof. This is the standard affine degree point-count bound, proved by induction on dimension using generic affine hyperplane sections. The parameter image is no larger than the incidence set. $\square$

For a threshold statement, use finitely many charts and agreement levels. Let $\mathfrak{X}_A$ be a chart cover of the declared aperiodic branch at exact agreement A . If each finite affine pivot chart has a univariate eliminant, set

$$B_{\text{ap}}^{\text{aff,elim}}(a) := \sum_{A=a}^n \sum_{X \in \mathfrak{X}_A} \sum_{h=0}^{t_A-1} \deg Q_{X,h}.$$

For projective slopes add the certified infinity contribution, and for degree- d curves replace the inner sum by the coefficient-pivot sum $\sum_{i=0}^d \sum_h \deg Q_{X,i,h}$. Exact root-union tables may replace these safe sums whenever duplicate parameters across charts are certified.

Definition 3.65 (aperiodic Hankel-packing certificate). An aperiodic Hankel-packing certificate for a row, a line or curve family, and an agreement threshold a consists of:

- (i) the tangent/common-code-line and quotient-image ledgers that have already been removed;
- (ii) for each exact agreement $A \in \{a, \dots, n\}$, either a regular minor certificate Δ_A or a declaration that the exact-agreement bucket is Hankel-singular;
- (iii) for every singular or near-capacity bucket, a finite constructible locator chart cover $\mathfrak{X}_A$ as in Definition 3.58;
- (iv) for affine lines, a list of pivot eliminants $Q_{X,h}(Z)$, or a failed-pivot record explaining which residual locus remains unsplit;
- (v) for projective slopes, the finite-patch eliminants plus the infinity-chart empty/nonempty certificate;
- (vi) for curves, coefficient-pivot eliminants $Q_{X,i,h}(\Gamma)$;
- (vii) a root-union table or a safe degree-sum table producing the declared aperiodic numerator $B_{\text{ap}}(a)$;
- (viii) if any chart uses the dimension-degree fallback, its degree, dimension, and an explicit statement that the resulting bound is only a fallback, not a sharp threshold certificate.

The verifier checks locator divisibility by P_D , chart coverage of the declared aperiodic complement, nonzero pivot tests, ideal membership of eliminants after saturation, and final root-count arithmetic.

Remark 3.66 (contributor workflow). For each row and exact agreement A , contributors should try the following in order. First test the overdetermined regular minors. If a nonzero minor is found, record its root set and remove roots already paid by tangent or quotient ledgers. If all maximal minors vanish identically, move the instance to the singular bucket. On the singular or near-capacity bucket, build locator charts, then split by affine pivots $B_h \neq 0$; for projective samplers also test the infinity chart $B = 0, A \neq 0$; for curves split by coefficient pivots $(V_i)_h \neq 0$. A chart is complete only when it has either a nonzero eliminant, a certified empty locus, or a named residual obstruction. Residual obstructions should be labelled as quotient-periodic, tangent/common-code-line, subfield/extension exceptional, or candidate new obstruction.

Remark 3.67 (status). Theorems 3.46, 3.59 and 3.63 and Corollary 3.62 are proved certificate theorems. They do not prove that all rows admit small aperiodic eliminants. They make the remaining MCA safe-side task concrete: after tangent and quotient branches are removed, construct small eliminants for the pivot charts, or identify the singular chart that prevents such an eliminant.

Lemma 3.68 (interleaving transfer). *For every linear code $C \subseteq \mathbb{F}^n$, every $s \geq 1$ and every $\delta \in (0, 1)$:*

$$\varepsilon_{\text{ca}}(C^{\equiv s}, \delta) \geq \varepsilon_{\text{ca}}(C, \delta), \quad \varepsilon_{\text{mca}}(C^{\equiv s}, \delta) \geq \varepsilon_{\text{mca}}(C, \delta).$$

Proof. For $h \in \mathbb{F}^n$ write $h^\Delta := (h, \dots, h) \in (\mathbb{F}^n)^s$. For any codeword tuple $\mathbf{c} = (c^{(1)}, \dots, c^{(s)}) \in C^{\equiv s}$, the columns where h^Δ and $\mathbf{c}$ agree are $\bigcap_i \{j : c_j^{(i)} = h_j\} \subseteq \{j : c_j^{(1)} = h_j\}$, so $\Delta_s(h^\Delta, C^{\equiv s}) \geq \Delta(h, C)$; taking $c^{(1)} = \dots = c^{(s)}$ gives equality: $\Delta_s(h^\Delta, C^{\equiv s}) = \Delta(h, C)$.

Fix (f_1, f_2) attaining $\varepsilon_{\text{ca}}(C, \delta)$ (resp. ε_{mca}) and consider the pair (f_1^Δ, f_2^Δ) for $C^{\equiv s}$, with the slope γ acting row-wise, so $f_1^\Delta + \gamma f_2^\Delta = (f_1 + \gamma f_2)^\Delta$. By the displayed equality, $\Delta_s((f_1 + \gamma f_2)^\Delta, C^{\equiv s}) \leq \delta$ iff $\Delta(f_1 + \gamma f_2, C) \leq \delta$. For the correlated condition, the pair (f_1^Δ, f_2^Δ) viewed as a $2s$ -row interleaved word has rows f_1 (s times) and f_2 (s times); by the same column argument its distance to $(C^{\equiv s})^{\equiv 2} = C^{\equiv 2s}$ equals $\Delta_2((f_1, f_2), C^{\equiv 2})$. For the MCA condition, a common set S explains f_1^Δ and f_2^Δ iff it explains f_1 and f_2 . Hence γ is CA-bad (resp. MCA-bad) for (f_1^Δ, f_2^Δ) iff it is so for (f_1, f_2) , and the maxima over pairs can only grow. $\square$

4 The main theorem

Theorem 4.1 (universal cap). *Let $B \subseteq \mathbb{F}$ be finite fields, let $q = |B|$, let $D \subseteq B^\times$ be a multiplicative coset of order n , and let k have $\rho = k/n \in (0, 1)$; set $C := \text{RS}[\mathbb{F}, D, k]$. Let $N \mid n$ satisfy $a := n/N \mid k$ and $(1 - \rho)N \geq 3$. If*

$$\binom{N}{\rho N + 2} \geq |B| \left(\frac{q}{k} + 1 \right), \quad (1)$$

then, writing

$$\delta_N := 1 - \rho - \frac{2}{N},$$

we have

$$\varepsilon_{\text{ca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{n}{q} \right)$$

for every integer-admissible radius

$$\delta_N \leq \delta \leq 1 - \rho - \frac{1}{n}.$$

In particular,

$$\varepsilon_{\text{mca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{n}{q} \right)$$

for every

$$\delta_N \leq \delta < 1 - \rho.$$

Consequently,

$$\delta_C^*(\varepsilon^*) \leq 1 - \rho - \frac{2}{N}$$

for every

$$\varepsilon^* \leq \frac{1}{2k} \left(1 - \frac{n}{q} \right).$$

Proof. Note first that $\rho N = k/a \in \mathbb{Z}$, $\ell_2 = \rho N + 2 \leq N - 1$ by $(1 - \rho)N \geq 3$, and $\delta_N = 1 - \rho - 2/N > 0$ for the same reason. Also $k + 1 \leq n$.

Fix

$$\delta \in [\delta_N, 1 - \rho - \frac{1}{n}].$$

Then

$$\lfloor \delta n \rfloor \leq n - k - 1,$$

so Theorem 2.6 is applicable.

Suppose toward a contradiction that

$$\varepsilon_{\text{ca}}(C, \delta) \leq \frac{1}{2k} \left(1 - \frac{n}{q}\right).$$

Apply Theorem 2.6 with $\eta = \frac{1}{2}$. Its hypothesis holds, so

$$\Lambda(C^+, \delta) \leq \left\lceil 2q \cdot \varepsilon_{\text{ca}}(C, \delta) \right\rceil \leq \left\lceil \frac{q-n}{k} \right\rceil < \frac{q-n}{k} + 1 \leq \frac{q}{k} + 1.$$

On the other hand, by Lemma 3.1(ii) and monotonicity of lists in the radius, since $\delta \geq \delta_N$,

$$\Lambda(C^+, \delta) \geq |\Lambda(C^+, \delta_N, u_z)| \geq \binom{N}{\ell_2} / |B| \geq \frac{q}{k} + 1$$

by (1). This is a contradiction. Hence

$$\varepsilon_{\text{ca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{n}{q}\right)$$

for every $\delta \in [\delta_N, 1 - \rho - 1/n]$.

In particular, the above holds at $\delta = \delta_N$. By Fact 2.4,

$$\varepsilon_{\text{mca}}(C, \delta_N) \geq \varepsilon_{\text{ca}}(C, \delta_N) > \frac{1}{2k} \left(1 - \frac{n}{q}\right).$$

By support-wise MCA monotonicity, Lemma 2.5, the same lower bound for $\varepsilon_{\text{mca}}(C, \delta)$ holds for every

$$\delta \in [\delta_N, 1 - \rho].$$

Thus no sub-capacity radius

$$\delta \geq 1 - \rho - \frac{2}{N}$$

is ε^* -admissible whenever

$$\varepsilon^* \leq \frac{1}{2k} \left(1 - \frac{n}{q}\right),$$

which proves the claimed bound on $\delta_C^*(\varepsilon^*)$. $\square$

Remark 4.2 (constants). Choosing $\eta = 1 - \theta$ in Theorem 2.6 improves the conclusion to $\varepsilon_{\text{ca}} > \frac{1-\theta}{k} (1 - \frac{n}{q})$ at the price of strengthening (1) to $\binom{N}{\rho N + 2} \geq |B| (\frac{q}{\theta k} + 1)$; given the hundreds of bits of slack in (1) throughout the challenge envelope (Table 1), the error constant is effectively $(1 - o(1))/k$. We fix $\eta = \frac{1}{2}$ for cleanliness.

Remark 4.3 (scope). No smoothness is assumed beyond the existence of a single divisor $N \mid n$ with $a = n/N \mid k$ and (1). The theorem applies verbatim to 2-adic, mixed-radix, or any other FFT-friendly multiplicative domain, and — via the subfield pigeonhole — to any extension field $\mathbb{F} \supseteq B$ with $D \subseteq B$.

The deep-point conversion is an integer-radius theorem with the condition $\lfloor \delta n \rfloor \leq n - k - 1$. Therefore the no-loss CA lower bound is stated for

$$\delta \leq 1 - \rho - \frac{1}{n}.$$

The MCA threshold cap is stronger in the range needed for the Proximity Prize: once the support-wise MCA lower bound is proved at

$$\delta_N = 1 - \rho - \frac{2}{N},$$

monotonicity of support-wise MCA extends it to every larger $\delta < 1 - \rho$.

5 Consequences for the grand MCA challenge

Corollary 5.1 (universal field-size cap for the challenge envelope). *Let $\rho \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\}$ and set*

$$N_\rho := 1024 \quad (\rho \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\}), \quad N_{1/16} := 2048.$$

Let $\mathbb{F}$ be any finite field with $q < 2^{256}$, let $B \subseteq \mathbb{F}$ be any subfield, let $D \subseteq B^\times$ be a multiplicative coset of order n with $N_\rho \mid n$, and let $k = \rho n \leq 2^{40}$. Then $C = \text{RS}[\mathbb{F}, D, k]$ satisfies

$$\varepsilon_{\text{mca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{n}{q}\right) \geq 2^{-86} \gg 2^{-128}$$

for every

$$\delta \in \left[1 - \rho - \frac{2}{N_\rho}, 1 - \rho\right).$$

Moreover, the same lower bound holds for $\varepsilon_{\text{ca}}(C, \delta)$ throughout the integer-admissible subinterval

$$\delta \in \left[1 - \rho - \frac{2}{N_\rho}, 1 - \rho - \frac{1}{n}\right].$$

If $q \geq 2n$, the lower bound improves to $\geq 2^{-42}$. In particular,

$$\delta_C^*(2^{-128}) \leq 1 - \rho - 2^{-9} \quad (\rho \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\}), \quad \delta_C^*(2^{-128}) \leq 1 - \rho - 2^{-10} \quad (\rho = \frac{1}{16}).$$

Proof. We verify the hypotheses of Theorem 4.1 with $N = N_\rho$. Divisibility: $a = n/N_\rho$ divides k because $k/a = \rho N_\rho \in \mathbb{Z}$ (equal to 512, 256, 128, 128 respectively); $(1 - \rho)N_\rho \geq 128 \geq 3$. For (1), the entropy bound $\binom{N}{\ell} \geq 2^{N H_2(\ell/N)}/(N+1)$ together with the conservative evaluations of H_2 recorded in Table 1 gives $\log_2 \binom{N_\rho}{\rho N_\rho + 2} \geq 552$ in all four cases, while

$$|B| \binom{\frac{q}{k} + 1}{k} \leq q \binom{\frac{q}{k} + 1}{k} \leq q^2 + q < 2^{512} + 2^{256} < 2^{513} \leq 2^{552},$$

so (1) holds with at least 39 bits to spare (and over 500 bits at rate $\frac{1}{2}$). Since $D \subseteq \mathbb{F}^\times$, we have $n \leq q - 1$; for fixed n , the function $1 - n/q$ is minimized at $q = n + 1$, hence $1 - n/q \geq 1/(n + 1)$. Also $n = k/\rho \leq 16k \leq 2^{44}$. Therefore

$$\frac{1}{2k} \left(1 - \frac{n}{q}\right) \geq \frac{1}{2k(n + 1)} > 2^{-86}.$$

If $q \geq 2n$ then $1 - n/q \geq 1/2$, giving the sharper $1/(4k) \geq 2^{-42}$. The gap is $2/N_\rho = 2^{-9}$ for $\rho \in \{1/2, 1/4, 1/8\}$ and $2/N_\rho = 2^{-10}$ for $\rho = 1/16$. Theorem 4.1 gives the MCA lower bound throughout the stated sub-capacity interval, and gives the CA lower bound on the integer-admissible subinterval ending at $1 - \rho - 1/n$. $\square$

ρ	N_ρ	$\ell_2 = \rho N_\rho + 2$	$H_2(\ell_2/N_\rho) \geq$	$\log_2 \binom{N_\rho}{\ell_2} \geq$	needed	gap $2/N_\rho$
1/2	1024	514	0.99998	1013	513	2^{-9}
1/4	1024	258	0.8143	823	513	2^{-9}
1/8	1024	130	0.5490	552	513	2^{-9}
1/16	2048	130	0.34105	687	513	2^{-10}

Table 1: Numerics for Corollary 5.1: $\log_2 \binom{N}{\ell} \geq NH_2(\ell/N) - \log_2(N+1)$, rounded down. The “needed” column is the universal upper bound $\log_2(|B|(q/k+1)) < 513$ valid throughout the envelope $q < 2^{256}$, $k \geq 1$.

Remark 5.2 (hypothesis $N_\rho \mid n$). For the 2-adic domains of the challenge this holds whenever $n \geq 2^{10}$ (resp. 2^{11}), i.e. for all but very small instances. For mixed-radix smooth n , replace N_ρ by any divisor $N \mid n$ of comparable size with $\rho N \in \mathbb{Z}$; the entropy count in Table 1 is insensitive to the exact value of N at fixed $\log_2 N$.

Remark 5.3 (two-tier answer to the grand challenge). Combining Corollary 5.1 with the grand-challenge cap of [Cho26b, Prop. “Grand-challenge cap”] (gap $\frac{1}{64}$, prime fields up to $\approx 2^{150}$, via value-set coverage) yields the current state of the negative side for smooth multiplicative domains:

$$\delta_C^*(2^{-128}) \leq \begin{cases} 1 - \rho - \frac{1}{64}, & \mathbb{F} \text{ prime, } q \lesssim 2^{150}, \\ 1 - \rho - 2^{-9}, & \rho \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\}, \text{ every } \mathbb{F}, q < 2^{256}, \\ 1 - \rho - 2^{-10}, & \rho = \frac{1}{16}, \text{ every } \mathbb{F}, q < 2^{256}. \end{cases}$$

Together with the positive results collected in [ABF26, Table 1], this pins the grand MCA threshold between the Johnson region and the explicit capacity-edge caps above, throughout the stated challenge envelope. The second and third tiers supersede the conditional cap of [Cho26b, Thm. “Universal cap”] ($\delta^* < 1 - \rho - 2^{-11}$ under $|\mathbb{F}| \geq 2n$), which was the previous state in this band.

Corollary 5.4 (deployed parameters: KoalaBear sextic). *Let $B = \mathbb{F}_p$ with $p = 2^{31} - 2^{24} + 1$, let $\mathbb{F} = \mathbb{F}_{p^6}$ (so $q = p^6 \approx 2^{185.93}$), let $D \subseteq B^\times$ be the subgroup of order $n = 2^{21}$, and let $k = 2^{20}$, $\rho = \frac{1}{2}$, as in [ABF26, §6.3]. Then*

$$\varepsilon_{\text{mca}}(\text{RS}[\mathbb{F}_{p^6}, D, 2^{20}], \delta) > (1 - 2^{-164}) 2^{-21} > 2^{-22}$$

for every

$$\delta \in \left[\frac{1}{2} - 2^{-7}, \frac{1}{2} \right).$$

Moreover, the same lower bound holds for ε_{ca} throughout the integer-admissible subinterval

$$\delta \in \left[\frac{1}{2} - 2^{-7}, \frac{1}{2} - 2^{-21} \right).$$

Proof. Apply Theorem 4.1 with $N = 256$, $a = 2^{13} \mid k$, $(1 - \rho)N = 128 \geq 3$, $\text{gap } 2/N = 2^{-7}$. Hypothesis (1): $\log_2 \binom{256}{130} \geq 256 \cdot H_2(130/256) - \log_2 257 \geq 256 \cdot 0.99982 - 8.01 > 247$, while $|B|(q/k+1) < 2^{31}(2^{166}+1) < 2^{198}$. The error bound is $\frac{1}{2k}(1 - n/q)$ with $n/q = 2^{21}/p^6 < 2^{-164}$. Theorem 4.1 gives the stated MCA bound on $[\frac{1}{2} - 2^{-7}, \frac{1}{2})$ and the stated CA bound on $[\frac{1}{2} - 2^{-7}, \frac{1}{2} - 2^{-21})$. $\square$

Corollary 5.5 (all interleaved rows of [ABF26, Tables 2–3]). *In the setting of Corollary 5.4, for each $s = 2^j$ with $0 \leq j \leq 12$ let $C_s := \text{RS}[\mathbb{F}_{p^6}, D_s, k_s]$ with $|D_s| = n_s = 2^{21}/s$ and $k_s = 2^{20}/s$ (rate $\frac{1}{2}$, $s n_s = 2^{21}$ as deployed). Then for every $\delta \in [\frac{1}{2} - 2^{-7}, \frac{1}{2})$,*

$$\varepsilon_{\text{mca}}(C_s^{\equiv s}, \delta) > \frac{s}{2^{21}}(1 - 2^{-164}) \geq 2^{-21}(1 - 2^{-164}).$$

Moreover, the same lower bound holds for $\varepsilon_{\text{ca}}(C_s^{\equiv s}, \delta)$ on the integer-admissible subinterval

$$\delta \in \left[\frac{1}{2} - 2^{-7}, \frac{1}{2} - \frac{1}{n_s}\right).$$

Every row of the deployed parameter family therefore fails the 2^{-128} target at gap 2^{-7} below capacity.

Proof. For each $s \leq 2^{12}$: $256 \mid n_s$ (as $n_s \geq 2^9$), $a_s = n_s/256 = 2^{13}/s$ divides $k_s = 2^{20}/s$, and $(1 - \rho) \cdot 256 = 128 \geq 3$. Hypothesis (1) for C_s : $|B|(q/k_s + 1) < 2^{31}(2^{166}s + 1) < 2^{198}s \leq 2^{210} \leq 2^{247} \leq \binom{256}{130}$. Theorem 4.1 gives the MCA lower bound for C_s throughout $[\frac{1}{2} - 2^{-7}, \frac{1}{2})$, and the CA lower bound throughout $[\frac{1}{2} - 2^{-7}, \frac{1}{2} - 1/n_s]$. Lemma 3.68 transfers the corresponding lower bounds to the s -interleaved code. $\square$

Proposition 5.6 (independent slacked variant via Theorem 2.16). *In the setting of Theorem 4.1, assume instead $(1 - \rho)N \geq 2$, $n \geq 3N$, and*

$$\binom{N}{\rho N + 1} \geq |B| \cdot q.$$

Then

$$\varepsilon_{\text{ca}}\left(C, 1 - \rho - \frac{1}{N} + \frac{2}{n}, 1 - \rho - \frac{1}{n}\right) \geq \frac{1}{2n}.$$

In particular, with $N = N_\rho$ as in Corollary 5.1 and $n \geq 3N_\rho$, every challenge-envelope instantiation satisfies this slacked CA lower bound, and $\frac{1}{2n} \geq 2^{-45} \gg 2^{-128}$.

Proof. Set $\delta_0 := 1 - \rho - \frac{1}{N}$. By $(1 - \rho)N \geq 2$, $\delta_0 \in (0, 1 - \rho)$. By $n \geq 3N$, we also have $\delta_0 + 2/n \leq 1 - \rho - 1/n$, so the parameters in Theorem 2.16 are admissible. By Lemma 3.1(i), some u_z carries at least $\binom{N}{\rho N + 1}/|B| \geq q$ codewords of C within radius δ_0 , so $\Lambda(C, \delta_0) \geq q$. The contrapositive of Theorem 2.16 gives

$$\varepsilon_{\text{ca}}\left(C, \delta_0 + \frac{2}{n}, 1 - \rho - \frac{1}{n}\right) \geq \frac{1}{2n}.$$

For the numerical claim, the worst case of $\log_2 \binom{N_\rho}{\rho N_\rho + 1}$ over the four rates is at $\rho = 1/8$, where the entropy bound used in Table 1 certifies $\log_2 \binom{1024}{129} \geq 1024 H_2(129/1024) - \log_2 1025 \geq 549$ (the exact value is ≈ 554.7); hence $\binom{N_\rho}{\rho N_\rho + 1} \geq 2^{549} > 2^{512} > |B|q$ throughout $q < 2^{256}$. Finally $n = k/\rho \leq 16k \leq 2^{44}$, so $1/(2n) \geq 2^{-45}$. $\square$

Remark 5.7. Proposition 5.6 is logically independent of Theorem 2.6 and yields a weaker error ($\frac{1}{2n}$ versus $\frac{1}{2k}$) in the *maximally slacked* form of correlated agreement — the internal radius sits at capacity minus $\frac{1}{n}$ — which is the weakest failure mode one can certify. That even this fails by 83 orders of magnitude (in bits) above the 2^{-128} target gives a separate fallback route through the imported BCHKS/ABF theorem.

6 Subfield confinement and the shape of certifying lines

The deployed instantiations run the protocol over an extension $\mathbb{F}/B$ while keeping the domain in the base: $D \subseteq B$. The subfield-confinement theorem of [Cho26b, Thm. “Subfield confinement”] pins down where bad slopes can live for lines whose words are base-rational: all support-wise MCA-bad slopes lie in B . The following lemma refines this by tracking two radii, so that it also covers the proximity-loss form of correlated agreement (Definition 2.1); part (b) is the statement of [Cho26b], reproved here for completeness in the present notation.

Lemma 6.1 (subfield confinement; refines [Cho26b]). *Let $B \subseteq \mathbb{F}$ be finite fields, $D \subseteq B^\times$, $C = \text{RS}[\mathbb{F}, D, k]$, and let $f, g \in B^n \subseteq \mathbb{F}^n$ be B -valued words. Then:*

- (a) *for any $0 \leq \delta_{\text{fld}} \leq \delta_{\text{int}} < 1$, every CA-bad slope for (f, g) at radii $(\delta_{\text{fld}}, \delta_{\text{int}})$ lies in B ;*
- (b) *for any $\delta \in (0, 1)$, every MCA-bad slope for (f, g) at radius δ lies in B .*

Consequently the bad-slope set of any B -valued pair has density at most $|B|/q$ in $\mathbb{F}$. The same holds for pairs $(\lambda f, \lambda g)$ with $\lambda \in \mathbb{F}^\times$ and f, g B -valued, by $\mathbb{F}$ -linearity of C .

Proof. Fix a B -basis $1 = \beta_0, \beta_1, \dots, \beta_{m-1}$ of $\mathbb{F}$. Every word $w \in \mathbb{F}^n$ decomposes uniquely as $w = \sum_i \beta_i w_i$ with $w_i \in B^n$, and every codeword $c \in \text{RS}[\mathbb{F}, D, k]$ decomposes as $c = \sum_i \beta_i c_i$ with $c_i \in \text{RS}[B, D, k]$: write the coefficient vector of the underlying polynomial in the basis and evaluate at $D \subseteq B$.

Let $z \in \mathbb{F} \setminus B$, say $z = \sum_i z_i \beta_i$ with $z_{i^*} \neq 0$ for some $i^* \geq 1$. We first record the support-wise consequence. Suppose that $c \in C$ agrees with $f + zg$ on some set $S \subseteq D$. Since $f_j, g_j \in B$, the basis components of $(f + zg)_j$ are $f_j + z_0 g_j$ (component 0) and $z_i g_j$ (components $i \geq 1$). Agreement on S holds componentwise, so on S :

$$f + z_0 g = c_0, \quad z_{i^*} g = c_{i^*}.$$

Hence g agrees on S with $z_{i^*}^{-1} c_{i^*} \in C$, and then $f = (f + z_0 g) - z_0 g$ agrees on S with $c_0 - z_0 z_{i^*}^{-1} c_{i^*} \in C$. Thus every support on which the line point $f + zg$ is explained by C also explains the pair (f, g) on that same support.

For CA, if $\Delta(f + zg, C) \leq \delta_{\text{fld}}$, choose such an S with $|S| \geq (1 - \delta_{\text{fld}})n$; the preceding paragraph gives $\Delta_2((f, g), C^{\equiv 2}) \leq \delta_{\text{fld}} \leq \delta_{\text{int}}$, so z is not CA-bad. For MCA, the preceding paragraph rules out every possible bad support S in the support-wise definition. Hence z is not MCA-bad either. $\square$

Corollary 6.2 (certifying lines over extensions are genuinely $\mathbb{F}$ -valued). *At the parameters of Corollary 5.4, $|B|/q = p^{-5} < 2^{-154}$. Hence any pair (f, g) whose CA- or MCA-bad-slope density at some radius*

$$\delta \in \left[\frac{1}{2} - 2^{-7}, \frac{1}{2} \right)$$

exceeds 2^{-154} contains a word that is not B -valued, even after removing a common scalar factor. In particular, any pair attaining the MCA lower bound $> 2^{-22}$ of Corollary 5.4 is genuinely $\mathbb{F}$ -valued. On the integer-admissible subinterval

$$\delta \in \left[\frac{1}{2} - 2^{-7}, \frac{1}{2} - 2^{-21} \right),$$

the same statement applies to pairs attaining the CA lower bound. All explicit failure witnesses currently known on these domains (the locator lines of [Cho26a, Cho26b] and the fiber words of

Lemma 3.1) are B -rational and are therefore inert here: the failure certified by Corollary 5.4 is fiber-borne and, at present, non-constructive.

Proof. Immediate from Lemma 6.1 and Corollary 5.4: a B -valued pair, up to common scalar, has CA- and MCA-bad-slope density at most $|B|/q = p^{-5} < 2^{-154} < 2^{-22}$. Thus any pair witnessing the deployed lower bound cannot be of this form. $\square$

Remark 6.3. Lemma 6.1 and Corollary 6.2 sharpen the division of labor between the two grand challenges of [ABF26] over deployed extensions: the *list* side does not deflate ($\text{RS}[B, D, k] \subseteq \text{RS}[\mathbb{F}, D, k]$, so all base-field list lower bounds, including Lemma 3.1, persist verbatim), while every sub- 2^{-128} MCA attack by explicit base-rational witnesses is killed by confinement — and yet MCA still fails at 2^{-22} . In earlier drafts this failure was only extracted non-constructively from Theorem 2.6; Corollaries 2.12 and 2.13 now give a constructive source whenever a large list is supplied: choose the simple pole in $\mathbb{F} \setminus B$ and print its value-bucket table. The deflation and decoupling corollaries of [Cho26b] record the same division at the no-loss radius, using the same corrected rounding $p^{-5} = 2^{-154.94\dots} < 2^{-154}$ as in Corollary 6.2. Exhibiting deployed-size bucket tables for such lines is Open Problem 8.1.

7 Prize-facing status and a completion program

The Proximity Prize asks for the largest admissible radius, not only for a near-capacity obstruction. Thus the present paper should be read as a one-sided theorem: it proves a universal unsafe band, while the remaining work is to prove a matching safe band below that cap, or to find additional obstruction floors that push the threshold even farther from capacity.

7.1 What is already settled by this paper

For the grand MCA challenge at target error $\varepsilon^* = 2^{-128}$, the unconditional output of Corollary 5.1 is the following negative certificate: for every challenge-envelope row satisfying the printed divisor hypothesis,

$$\delta \geq 1 - \rho - 2^{-9} \quad (\rho \in \{1/2, 1/4, 1/8\})$$

and

$$\delta \geq 1 - \rho - 2^{-10} \quad (\rho = 1/16)$$

are impossible MCA radii. Equivalently, any full determination of $\delta_C^*(2^{-128})$ for these smooth multiplicative rows must search below these gaps. This conclusion uses only the self-contained deep-point conversion, the quotient-fiber pigeonhole, and support-wise MCA monotonicity; it no longer rests on importing the Crites–Stewart conversion.

The theorem also distinguishes two different grades of information. First, the *threshold grade* is already enough for the prize inequality: the certified error is $\gg 2^{-128}$. Second, the *failure-profile grade* is still weak: the present proof gives error of order $1/k$, not error $1 - o(1)$. Closing the error-one profile at fixed gaps such as $1/64$ is a different, stronger problem. It would improve our understanding of the failure landscape, but is not needed to obtain the stated upper bound on $\delta_C^*(2^{-128})$.

7.2 Integer-staircase formulation

Let

$$a(\delta) := \lceil (1 - \delta)n \rceil, \quad r(\delta) := n - a(\delta)$$

be the agreement and integer-radius forms of the same radius. For a fixed finite-slope line field F_{line} , write $q_{\text{line}} = |F_{\text{line}}|$ and let

$$B_{\text{mca}}(a)$$

be the exact maximum number of MCA-bad line parameters at agreement at least a , for the line family used in the challenge. Then

$$\varepsilon_{\text{mca}}(C, 1 - a/n) = B_{\text{mca}}(a)/q_{\text{line}}$$

under the closed finite-slope convention. For projective-slope or curve samplers, the same staircase formulation applies after replacing q_{line} and B_{mca} by the corresponding sampler denominator and numerator. Thus the MCA prize problem is an integer staircase problem: find the first agreement a for which

$$B_{\text{mca}}(a) \leq 2^{-128} q_{\text{line}}.$$

The present paper supplies lower bounds on $B_{\text{mca}}(a)$ near capacity by routing quotient-fiber list mass through deep points.

Proposition 7.1 (one-step threshold certificate). *Fix a row C and a denominator q_{line} . Suppose that for some agreement a_0 one has a lower certificate*

$$B_{\text{mca}}(a_0) > 2^{-128} q_{\text{line}}$$

and an upper theorem

$$B_{\text{mca}}(a_0 + 1) \leq 2^{-128} q_{\text{line}}.$$

Then, under the closed integer-radius convention, radius $1 - a_0/n$ is unsafe and radius $1 - (a_0 + 1)/n$ is safe. As a real-radius supremum, the threshold is $1 - a_0/n$ and is not attained if the closed ball at that endpoint is counted.

Proof. The function $B_{\text{mca}}(a)$ is nonincreasing in a , since raising the required agreement only removes witnesses. At the closed radius $1 - a/n$, the MCA error is $B_{\text{mca}}(a)/q_{\text{line}}$. The two displayed inequalities are therefore exactly the unsafe/safe transition at consecutive integer agreement levels. $\square$

7.3 Conditional MCA completion theorem

The following theorem is a deliberately explicit proof blueprint. Each hypothesis is a concrete mathematical task, not a black-box “large field” assumption.

Theorem 7.2 (conditional MCA threshold theorem). *Fix a smooth multiplicative row $C = \text{RS}[F, D, k]$, a line field F_{line} , and a target $\varepsilon^* = 2^{-128}$. Suppose the following four inputs are proved for this row, uniformly in the received line:*

- (i) **Quotient-profile lower and upper ledger.** *For every divisor $c \mid n$ and every agreement $a = mc + s$ with $0 \leq s < c$, the quotient-periodic bad slopes, including remainder supports consisting of m full fibers plus s residual points, are counted with overlaps across divisors accounted for. Write the resulting divisor-lattice upper numerator as $B_Q^{\text{all}}(a)$; individual divisor contributions $B_Q(a; c)$ may also be printed, but the upper ledger must count their union or a certified safe sum rather than only their maximum.*

- (ii) **Aperiodic residue-line packing.** After removing tangent and quotient-periodic locators, the Hankel/residue-line incidence variety has at most $B_{\text{ap}}(a)$ bad line parameters.
- (iii) **Extension-line accounting.** If F_{line} is a proper extension of the field generated by D , either all genuinely extension-valued line witnesses are included in $B_{\text{ap}}(a)$, or a separate extension term $B_{\text{ext}}(a)$ is proved.
- (iv) **No missing line family.** The line family in the theorem is the same finite-slope or projective-slope family used by the challenge or protocol under consideration.

Assume moreover that the resulting upper numerator

$$B^+(a) := B_{\text{tan}}(a) + B_Q^{\text{all}}(a) + B_{\text{ap}}(a) + B_{\text{ext}}(a)$$

and a corresponding lower numerator $B^-(a)$ satisfy

$$B^-(a_0) > \varepsilon^* q_{\text{line}}, \quad B^+(a_0 + 1) \leq \varepsilon^* q_{\text{line}}$$

for some a_0 . Then $\delta_C^*(\varepsilon^*)$ is determined at the one-step accuracy of Proposition 7.1.

Proof. The four hypotheses classify every possible MCA-bad line parameter into one of four buckets: tangent, quotient-periodic, aperiodic, or genuinely extension-line. The quotient bucket is counted by the divisor-lattice union ledger B_Q^{all} , so overlaps or simultaneous quotient descriptions cannot invalidate the upper bound. The displayed $B^+(a)$ is therefore an upper bound for $B_{\text{mca}}(a)$, while $B^-(a)$ is a certified lower bound. The conclusion is exactly Proposition 7.1. $\square$

The present paper supplies part of the lower numerator $B^-(a)$ in the near-capacity band: quotient-fiber list mass forces MCA mass above the printed universal caps. It does not prove the aperiodic upper input (ii), and therefore does not by itself determine the exact staircase.

7.4 Conditional interleaved-list completion theorem

The grand list-decoding challenge asks for the largest δ such that an interleaved list, divided by the field used for the relevant challenge or list ledger, is below 2^{-128} . In the simplest same-field form this is

$$|\Lambda(C^{\equiv m}, \delta)| \leq 2^{-128} |F|.$$

The quotient-fiber construction in this paper already gives lower floors for ordinary lists, and ordinary lists embed into interleaved lists by placing the same received word in one row and zero words in the others. On the MCA side, Section 3.4 supplies a certified eliminant route for aperiodic Hankel packing, but it does not prove that the needed eliminants always exist. The missing positive list direction is an arbitrary-word locator-fiber theorem.

Theorem 7.3 (conditional interleaved-list threshold theorem). *Fix an interleaving arity m . Suppose that for every received word U and agreement $a = k + \sigma$ one has a base-code list bound of the form*

$$|\Lambda(C, 1 - a/n, U)| \leq L_Q(a, U) + L_{\text{ap}}(a),$$

where L_Q is an explicit quotient-profile list contribution and L_{ap} is a uniform aperiodic locator-fiber bound. Suppose also that the same bound is available at the higher agreement levels that occur in the codegree recursion, starting with $2a - k$. Then the interleaved-list threshold is obtained by applying the codegree recursion row by row and comparing the resulting numerator to $2^{-128} |F|$.

Proof. For two rows, split the second-row list according to whether its agreement set has size below or above $2a - k$. Below $2a - k$, two first-row codewords would agree on more than k positions and hence are equal; above $2a - k$, use the base-code list bound at agreement $2a - k$. Iterating gives the stated m -row recursion. This is the same codegree mechanism used in the experimental ledger; the only missing input is the arbitrary-word locator-fiber bound. $\square$

7.5 Concrete proof obligations

The fastest route from the present cap to a full prize solution is to turn the following proof obligations into theorems. The lower-floor half of T1 is now supplied in scanner-ready form by Lemma 3.7 and Theorem 3.8: every quotient-plus-remainder agreement $a = mc + s$ has a generated-field prefix-fiber floor, an image-size fallback, an ambient-box fallback, and a converted CA/MCA lower numerator. The support-family upper half is exact at the divisor-lattice level by Theorem 3.24: overlaps between quotient descriptions at different divisor scales are counted by a common-block coefficient extraction rather than by a raw safe sum. The distinct-parameter branch image is exact inside every declared quotient family by the lcm ledger of Theorem 3.36 and Corollaries 3.37 and 3.39; duplicate finite slopes, projective slopes, and finite curve parameters are coalesced before the numerator is printed. T2 is now also partially upgraded on its regular overdetermined branch by Theorems 3.48 and 3.55 and Corollary 3.51: instead of choosing one regular Hankel minor, the scanner takes the gcd of all maximal minors, then an lcm over exact agreements, and finally removes exact quotient/tangent paid roots. What remains open is the genuinely structural safe-side theorem: proving that every bad parameter in the intended near-capacity regime either belongs to the tangent or quotient image, or is bounded by the pivot-chart aperiodic and extension ledgers. The underdetermined and rank-drop singular Hankel buckets remain the named obstruction for T2. T5 is partially upgraded as well: finite-parameter curve and projective-line quotient branches have exact gcd/lcm image ledgers, and finite-parameter curve regular rank-drop branches have canonical maximal-minor gcd ledgers, while the aperiodic curve and projective pivot charts still require eliminant certificates.

T1. Remainder quotient-profile theorem. Generalize the fiber lemma from full quotient-fiber unions to all agreements $a = mc + s$. The target is an exact lower and upper quotient ledger over the divisor lattice of D . Version 10 supplies the scanner-ready lower half via heaviest prefix fibers and the support-union upper half via the exact divisor-block coefficient formula; inside a declared quotient family, the gcd/lcm image ledger also coalesces duplicate finite, projective, and curve parameters. What remains is the structural exhaustion theorem saying that the declared quotient family captures all quotient-periodic bad parameters in the intended regime, with the subfield denominator $|B|$ printed separately from the line denominator q .

T2. Aperiodic Hankel-packing theorem. Use the syndrome/Hankel normal form of [Cho26b] to prove that, after quotient-periodic and tangent locators are removed, the number of remaining bad line parameters is at most $B_{\text{ap}}(a)$. The regular overdetermined nonsingular branch is now canonical by Theorem 3.48 and Corollary 3.51: a scanner takes the gcd of all nonzero maximal Hankel minors and then an lcm over exact agreement levels, rather than choosing one minor. The remaining proof problem is concentrated in the underdetermined and rank-drop singular buckets: construct small pivot-chart eliminants for them, or classify their residual components as quotient, tangent, extension-field exceptional, or genuinely new.

- T3. Extension-line lift or counterexample.** For $B \subset F$ and $D \subset B$, either prove that genuinely F -valued lines are covered by the aperiodic ledger with a stable numerator, or exhibit explicit extension-valued counterexamples. Corollary 6.2 shows that deployed failures certified by this paper must be genuinely extension-valued; Corollaries 2.12 and 2.13 now provide an explicit lower-side mechanism from any large list by taking the pole in $F \setminus B$. The remaining safe-side problem is to classify all such extension-valued lines or charge them to the aperiodic ledger.
- T4. Arbitrary-word locator-fiber theorem.** Prove the base-code list bound needed by Theorem 7.3, first for monomial-prefix received words and quotient-free dimensions, then for arbitrary received words with quotient cores budgeted.
- T5. Curve and projective-line ledgers.** Extend the affine-line Hankel classification to projective slopes and to finite-parameter power curves $f_0 + \gamma f_1 + \cdots + \gamma^d f_d$. Version 10 gives exact gcd/lcm ledgers for projective and finite-curve quotient branches and canonical rank-drop gcfs for regular overdetermined finite-curve buckets. The remaining work is the aperiodic curve and projective pivot atlas, including exceptional vanishing-vector strata.
- T6. Certificate scanner.** Implement a finite staircase scanner that, for each challenge row, prints tangent, quotient, generated-field entropy, aperiodic, extension, curve, and interleaved-list numerators at every agreement a . The desired output is an interval

$$a_{\text{unsafe}} < a_C^* \leq a_{\text{safe}}$$

that shrinks to one step when Theorem 7.2 or Theorem 7.3 applies.

- T7. Formal finite core.** Formalize the self-contained deep-point conversion, the quotient-fiber pigeonhole, support-wise MCA monotonicity, and the integer staircase lemma. These are small enough for proof-assistant checking and directly strengthen a prize submission.

7.6 Recommended submission posture

As a standalone prize-facing submission, this paper should claim a significant partial result: a universal negative band for the grand MCA challenge over smooth multiplicative rows in the stated field-size envelope. It should not claim the full prize threshold, a positive theorem below the cap, protocol soundness, or a curve-MCA theorem. The full prize claim should be made only after the conditional ledgers above are replaced by theorems and the staircase scanner produces a one-step unsafe/safe transition for the relevant rows.

8 Discussion

Values versus fibers. Above the per-rate value-set reach of [Cho26b] (roughly 2^{150} – 2^{162}), the value route — showing that the slope multiset $e_1(\ell^\wedge Q)$ covers a 2^{-128} fraction of $\mathbb{F}$ at a fixed large prime — runs into the per-fiber collision problem ([Cho26b, Rem. “The remaining band”]): characteristic-zero locator values can collide arbitrarily badly under reduction at a worst-case prime. [Cho26b] therefore closed the band at the δ^* grade only conditionally and with weaker constants — error 2^{-42} at gap 2^{-11} , assuming $|\mathbb{F}| \geq 2n$, via quotient-core lists and the same conversion Theorem 2.6 — leaving the *error-one* grade open. The present route also never counts values, and improves each parameter of that cap: pigeonhole guarantees one

heavy fiber regardless of how the values $e_1(A)$ distribute, Theorem 2.6 converts heavy fibers into correlated-agreement error directly, and the subfield pigeonhole carries the construction to extension fields with no condition relating n and $|\mathbb{F}|$. The cost relative to the error-one results of [Cho26a, Cho26b] below 2^{150} is quantitative: error $\approx \frac{1}{2k}$ instead of $1 - o(1)$, and gap 2^{-9} (uniformly 2^{-10} across all four rates) instead of $\frac{1}{64}$. For the prize question — is $\varepsilon_{\text{mca}} \leq 2^{-128}$ achievable above the stated gaps? — this cost is immaterial.

Limits of the method. Hypothesis (1) needs $N H_2(\rho) \gtrsim 2 \log_2 q - \log_2 k$, so the smallest certifiable gap is

$$\frac{2}{N} \approx \frac{2 H_2(\rho)}{2 \log_2 q - \log_2 k} \approx \frac{H_2(\rho)}{\log_2 q},$$

and the certified error is capped at $\approx \frac{1}{k}$ by the shape of Theorem 2.6 (any $\eta < 1$). Since q enters only as the size of the field from which the line challenge γ is drawn, this is also a protocol-facing floor: enlarging the challenge field cannot push a deep-point-conversion failure below gap $\approx H_2(\rho) / \log_2 q_{\text{line}}$, which quantitatively bounds the “larger challenge field” fallback in certificate frameworks. The construction also needs $N \leq n$, i.e. $\log_2 q \lesssim H_2(\rho) n / 2$; this covers the entire cryptographic regime $\text{poly}(n) \leq q \leq 2^{o(n)}$, but for $q \geq 2^{H_2(\rho) n / 2}$ the method is vacuous and the value-route caps of [Cho26a, Cho26b] are the only ones known. Within the band, the *error-one* question at gap $\frac{1}{64}$ (Open Problem 8.2) remains exactly as open as before: the present result bounds δ^* , not the failure profile near capacity.

Comparison with the cyclotomic sieve. The sieve of [Cho26a] produces, for infinitely many primes, failures of error $(\log p)^{-6}$ at gap $\Theta(1 / \log p)$, but its prime set is given by Siegel–Walfisz and is ineffective. Corollary 5.1 trades error strength for universality: every field below 2^{256} , fully effective, and elementary given Theorem 2.6. The two results triangulate the failure landscape from different directions.

Pressure on the packing conjecture. By the exact normal form of [Cho26b], $\varepsilon_{\text{mca}}(C, \delta) = \max_t \Lambda_t^{\text{NC}}(\delta) / q$, where Λ^{NC} is a noncontained residue-line packing number. Corollary 5.4 therefore forces $\max_t \Lambda^{\text{NC}} \geq (q - n) / 2k \approx 2^{164}$ at gap 2^{-7} on the deployed sextic — superpolynomially many noncontained residue lines at $n = 2^{21}$. This puts no direct pressure on the final MCA conjecture of [Cho26b, Conj. “Final floor- and quotient-corrected MCA asymptotic”]: at gap 2^{-7} both of its corrected-reserve hypotheses fail ($\sigma = 2^{14}$ lies below $Cn / \log_2 n \approx 2^{16.6}$, and $\sigma \log_2 q_{\text{gen}} = 2^{14} \cdot 31 \approx 2^{19}$ is far below $\log_2 \binom{n}{k+\sigma} \approx 2^{21}$), so the conjectured ceiling of the form $(n^{1+o(1)} + 2^{(\beta/H_2) Q_{H_2}(\eta)}) / q$ is simply not asserted there — the packing mass extracted through Theorem 2.6 lives in the region the framework already marks as below-floor. The genuine structural question this note raises for the framework of [Cho26b] is whether conversion-extracted fiber mass persists *above* the corrected reserve: whether, at reserves where the conjecture does apply, the fiber route still produces packings beyond the $n^{1+o(1)}$ aperiodic term plus the quotient-periodic term, which would force an additional fiber-borne term into the conjectured ceiling. This is open.

Verification caveat. The main cap no longer depends on importing the Crites–Stewart theorem: Theorem 2.6 proves the needed deep-point list-to-CA conversion directly, including the integer-radius condition $f \leq n - k - 1$. Theorem 2.16 remains a separate slacked fallback imported

through the BCHKS/ABF route, and readers using that fallback should consult the underlying source directly. The main composition is deliberately shallow: it uses the self-contained conversion, the locator-fiber pigeonhole, and support-wise MCA monotonicity.

Open problems.

Open Problem 8.1. Exhibit explicit pairs (f, g) over $\mathbb{F}_{p^6}$ (KoalaBear sextic, parameters of Corollary 5.4) with MCA-bad-slope density $> 2^{-22}$ at gap 2^{-7} . On the integer-admissible subinterval, ask for the corresponding CA-bad-slope density as well. By Corollary 6.2 such pairs exist and are not B -rational; the natural candidates are residue-line normal forms of [Cho26b] with denominator $E \in \mathbb{F}[X]$ not defined over B .

Open Problem 8.2. Error one in the band: does $\varepsilon_{\text{mca}}(C, 1 - \rho - \frac{1}{64}) = 1 - o(1)$ hold for all primes $2^{150} \leq p \leq 2^{256}$? This is the per-fiber collision problem proper; Corollary 5.1 caps δ^* there but says nothing above error $\approx \frac{1}{2k}$.

Open Problem 8.3. Improve the threshold of Theorem 2.6: any strengthening of the hypothesis range $\varepsilon_{\text{ca}} \leq \eta/k$ toward $\varepsilon_{\text{ca}} \leq \eta \cdot \text{polylog}(q)/k$, or of the list conclusion, immediately tightens the certified error of Theorem 4.1 toward $1/\text{poly log}$; the gap, by contrast, is governed by the binomial in (1) and would need a denser fiber construction to shrink.

Open Problem 8.4. Beyond multiplicative smoothness: Lemma 3.1 uses only that D is a union of complete fibers of a low-degree map with lacunary power structure. Carry the composition to isogeny-smooth domains — in particular the circle-group domains of Circle STARKs over $p = 2^{31} - 1$, where Chebyshev locators $\prod_b (T_{2^j}(x) - x_b)$ play the role of $\prod_b (X^a - b)$ — to obtain the same universal cap for that deployed family.

Acknowledgements. This note builds on the Crites–Stewart conversion perspective [CS25], the BCHKS theorem [BCHKS25], and the locator framework of [Cho26a, Cho26b], following the synthesis of Arnon–Boneh–Fenzi [ABF26]; it proves the main conversion needed here directly, sharpens the conditional universal-cap theorem, and refines the subfield-confinement theorem of [Cho26b].

References

- [PP26] The Proximity Prize. The Proximity Prize: \$1M Reed–Solomon Challenge. Preliminary version, 2026. <https://proximityprize.org/>.
- [ABF26] G. Arnon, D. Boneh, and G. Fenzi. Open problems in list decoding and correlated agreement. IACR Cryptology ePrint Archive, Report 2026/680, 2026. <https://eprint.iacr.org/2026/680>.
- [BCHKS25] E. Ben-Sasson, D. Carmon, U. Haböck, S. Kopparty, and S. Saraf. On proximity gaps for Reed–Solomon codes. IACR Cryptology ePrint Archive, Report 2025/2055, 2025. <https://eprint.iacr.org/2025/2055>.
- [Cho26a] P. Chojecki. Capacity-edge obstructions to Reed–Solomon mutual correlated agreement over smooth multiplicative domains. Manuscript, 2026.

- [Cho26b] P. Chojeki. Slack, quotient cores, and the entropy gap for smooth-domain Reed–Solomon codes: list fibers, cyclotomic rigidity, Galois-amplified collisions, and the corrected slack mutual-correlated-agreement theory. Manuscript, merged corrected edition, 2026.
- [CS25] E. Crites and A. Stewart. On Reed–Solomon proximity gaps conjectures. IACR Cryptology ePrint Archive, Report 2025/2046, 2025. <https://eprint.iacr.org/2025/2046>.